

PORTFOLIO *of Practice*

LZAD7003

Practice-based Research

A Co-design Practice of

Embroidered Body Data

with Rural Women in Lanzhou

Student: Zhang Yuki

ID: 220250941940

Supervisor: Thomas Jansen

Date: July 5, 2026

目录
CONTENTS

REFLECTIVE JOURNAL
Record the key decisions, obstacles encountered, successful experiences, and insights gained during the one-week collaborative design process.

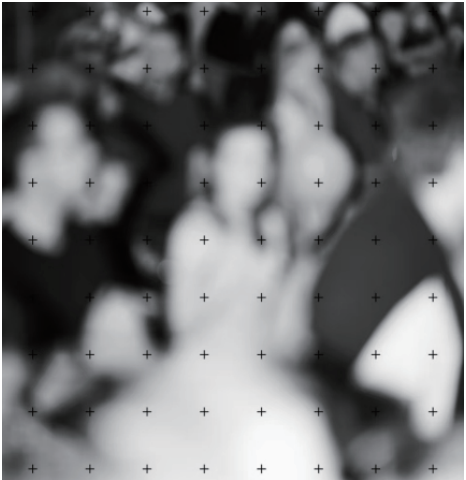
04-07

ANNOTATED BIBLIOGRAPHY

08-13

LITERATURE ASSIGNMENT
Read Tim Ingold, *Making*, Chapter 1 "Knowing from the Inside." Write a 500-word discussion explaining what Ingold means by 'knowing from the inside.'

14-15



ARTEFACT & REFLECTION
Use a completed creative work to discuss what specific actions you want to take to gather information about the work and its creation process.

20-27

RESEARCH QUESTIONS
Research questions explored through practice-based studies.

28-29

PBR PROPOSAL OUTLINE
Practical Research Proposal Outline

30-33

PAPER
Final paper

34-41

RESEARCH METHODS IN PRACTICE
Use research methods (observation/ interviews/questionnaires) to study specific aspects of the project.

16-19



Portfolio Introduction
Course: LZAD7003 Practice-based Research
Theme: Reciprocal Design Research on Marginalised Communities and Cultural Heritage – A Co-design Practice of Embroidered Body Data with Rural Women in Lanzhou
Student: Yuki
Submission Date: 5 July 2026

作品集说明课程：LZAD7003 实践型研究 (Practice-based Research)
主题：边缘群体与文化遗产的互惠设计研究——兰州农村女性身体数据刺绣共同设计实践
学生：雨琦
提交日期：2026 年 7 月 5 日

THE NATURE OF THIS PORTFOLIO
This portfolio is the final submission for the LZAD7003 Practice-based Research course, comprising all eight individual assessment components. Together, these components constitute a complete research cycle: from entering the field and recording reflections, to systematically reviewing literature and designing research methods, to producing a digital artefact and publicly presenting the research outcomes.

Unlike a conventional single-paper thesis, the portfolio format requires the researcher to exhibit the full folds of the research process—not only the "conclusions," but every step of "how those conclusions were reached": including moments of being corrected by participants, moments of confusion when methods failed, and moments of recalibration when technical solutions encountered resistance. These are the very elements typically filtered out by the smooth narratives of academic papers, but they are preserved and scrutinised in this portfolio. If a paper shows the "front side" of research, the portfolio shows its entirety—including its creases, folds, and unfinished edges.

Reflective Journal: Fieldnotes from a Co-design Project on Embroidered Body Data with Rural Women in Yuzhong County, Lanzhou
Researcher: Yuki
Field site: Yuzhong County, Lanzhou, Gansu Province
Dates: 10–17 June 2026
Theme: Reciprocal Design Research on Marginalised Communities and Cultural Heritage

反思日志：兰州榆中县农村女性身体数据刺绣共同设计田野笔记
田野地点：甘肃省兰州市榆中县
主题：边缘群体与文化遗产的互惠设计研究

研究者：雨琦

时间：2026 年 6 月 10 日—6 月 17 日

DAY 1: ENTERING THE FIELD – THE FAILURE OF PRECONCEPTIONS

Today was the first day of embroidering the individual pain panels. I call this practice "data embroidery"—a term I coined myself, rather than borrowing from any existing academic concept. Each woman had kept a pain diary over the previous five days and now translated that data into embroidery.

I observed Li Guilan embroidering the lotus motif representing her lower back, using extremely dense French knots to fill the flower centre. After about twenty minutes, she stopped and shook her hand. I asked what was wrong. She said: "My hand is sore. These knots take too much effort—I keep squeezing the needle tight." After a moment she added: "But it reminds me of that day in the field, bending down to pick vegetables—my lower back was sore in the same way."

She did not say "this reminds me of the pain"; she said "my lower back was sore in the same way"—the two experiences were the same bodily feeling for her. Ingold says "making is knowing," and at that moment I truly understood it. Coding is not using symbols to "represent" an experience external to it; it is re-activating that experience through the physical resistance of needle and thread passing through fabric. Her embroidered panel is not a record of pain but a re-enactment of pain. The density of stitches is not just data; it is the material trace of her fatigue from bending over in the vegetable field that afternoon.

I had previously understood "data embroidery" as "translating data into embroidery"—still a binary thinking. Today I saw that embroidery itself is the process of data generation. It is not that data comes first and embroidery follows; rather, data is generated in the act of embroidering.

DAY 2: THE CLASH OF STANDARDISATION AND THE TENSION IN THE ROOM

Today I tried a different approach. I did not mention the "three-level classification" again. Instead, I asked each woman to embroider a palm-sized square using whichever stitch she was most comfortable with—any pattern she liked. I wanted to observe their actual practice to understand what "sparse" and "dense" meant under their own hands.

The results confirmed yesterday's insight. Wang Xia (aged 32) embroidered a "dense" section that was almost as tight as Grandma Liu's (aged 67) "medium." When they placed their pieces side by side, they both laughed. Grandma Liu said: "When I was young, my hand was tighter too. Now I'm old, I can't grip the needle as firmly." She attributed the difference in stitch density to bodily ageing—this was not a technical error but a trace of bodily history.

But the atmosphere became subtle. When I suggested that "each of us could embroider a small sample of her own 'sparse,' 'medium,' and 'dense' as a personal baseline," several women stopped stitching and exchanged glances. Later, Li Guilan privately told me: "They're afraid that if their embroidery doesn't look good, you'll take it back and compare it with others." I barely slept that night. I thought I was offering a flexible solution, but to them, it still felt like a hidden form of evaluation and comparison. Their concern was not technical difficulty but the fear of being the one who "embroidered wrong."

The concept of "personal calibration" respects differences, but in implementation, it still carries the power pressure of "being watched and compared." I underestimated the existing social dynamics in the field—the women's long-term relationships with each other were far more complex than my relationship with them.

NO.1



Village woman's photography photo

Village woman's photography photo



REFLECTIVE JOURNAL

DAY 3: THE EMERGENCE OF METAPHOR – FROM ABSTRACT SCHEMES TO CONCRETE LANGUAGE

Today I made no structured arrangements. I said: "Today, don't embroider for the project—embroider whatever you want. I'll just watch."

Li Guilan was embroidering a maple leaf, filling the veins with dense French knots, the threads packed so tightly they seemed to crowd each other. I asked her: "How long are you going to spend on this leaf?" She answered, not directly: "Look at these knots, packed so tightly that the road is blocked, you can't move. Just like my knee—two days before it rains, I want to walk but can't."

Other women gradually joined the discussion. Wang Xia said: "French knots tire your hand. If you do too many, your fingers hurt—so it must mean the pain lasts long." Grandma Liu said: "Running stitch moves fast, no dragging—like when nothing's bothering you."

This was a process where an embodied metaphor emerged naturally from bodily experience. I did not guide it; I just stopped and listened. What I learned today was not "which metaphor can be used for communication," but: only when I stop framing my questions through my own schemas does their own language surface.

My earlier "interviews" were essentially translation—translating my questions into words they could understand, then translating their answers back into my theoretical framework. Today, the most productive thing I did was to stop translating and let them speak through their own needles and threads.

DAY 4: BEING CORRECTED IN MY "UNDERSTANDING"

I must record what happened today. It made me deeply uncomfortable, but it is extremely important. Last night, while organising my fieldnotes, I extensively analysed and theorised Li Guilan's metaphor of "can't move." I interpreted it as a "spatial projection of bodily-temporal experience," fitting the framework of ontological metaphor by Lakoff and Johnson, and planned to feed this analysis back to them today—with an attitude of "see, your expressions have great theoretical value."

In the afternoon, I asked Li Minghui to translate my interpretation into dialect for Li Guilan. The gist was: "We think your 'can't move' turns time into space—it's a sophisticated mode of cognition." Li Guilan paused for a few seconds, then smiled. She said: "You teachers really know how to think. I didn't think that much. When I embroider those knots, my hand holds the needle for a long time, and it gets sore, so I don't want to move. It's the same as when pain lasts long and you don't want to move." She gently dismantled the theoretical edifice I had built. She was not making a cognitive-linguistic "metaphorical mapping"—she was making a more concrete, more straightforward analogy: the hand fatigue from needlework reminded her of the fatigue from bodily pain. The two experiences share the same bodily feeling of "being tired and not wanting to move"—not an abstract conceptual operation. At that moment, I felt a mixture of embarrassment from being challenged and a deeper vigilance: to what extent was I repackaging their experience in academic language and calling that "understanding"? Is this not another form of hermeneutical injustice? I had not negated her words, but I had covered them with a louder theoretical voice.

A researcher's "analysis," if not validated by participants, risks becoming a form of epistemic violence. I must prioritise "how they understand their own metaphors" over "how I analyse them."

DAY 5: GENERATIONAL DIVERGENCE – THE LIMITS OF METAPHOR

Today we discussed colour coding. I proposed using a colour gradient to represent pain intensity and showed a spectrum ranging from light blue to deep red as a starting point for discussion.

The disagreement did not occur between me and them, but among themselves. Wang Xia said: "The lightest pain feels like wind passing by, a bit cool—I'd use light blue." Grandma Liu immediately disagreed: "When I was young, pain was burning—even the lightest was like fire. You say blue is pain—I say no." The two stood their ground, and other women split along generational lines—those under forty tended to accept cool colours, while those over fifty mostly insisted that pain must be warm-toned.

I tried to mediate by suggesting "a middle colour that everyone can accept," but Wang Xia asked: "That's neither blue nor red—what colour is it then?" She asked a valid question. Finding a middle ground meant cancelling out each person's experience. I realised that this disagreement could not be resolved through consensus because its root was not "different understandings of colours" but "different frameworks for sensing pain itself." Grandma Liu had spent her youth labouring in the fields—her acute physical injuries left a long-lasting imprint on her pain memory; Wang Xia worked mostly with light household tasks, her pain being chronic fatigue—the two sensory baselines were fundamentally different.

In the end, we did not unify the colour scheme. Instead, each person used her own colour chart in the individual calibration stage. It was a compromise, but I believe it was more honest than forced consensus.

Metaphor is not a panacea. When the source domain (temperature sensation, colour association) itself varies across generations and bodily histories, the same metaphor cannot carry shared meaning. Designers should not force such rifts to close but should preserve individual exits for expression.

DAY 6: NEEDLE AND THREAD AS A CHANNEL FOR BODILY MEMORY

Today we began embroidering the individual pain panels. I call this practice—turning pain diaries into needlework—"data embroidery." It is a term I coined myself, not borrowed from any established academic concept. What I mean by it is that embroidery is not an external translation of data; rather, it is the very generation of data. Each woman had kept a pain diary over the previous five days, and now they were to "embroider" those records.

I observed Li Guilan embroidering the lotus motif that represented her lower back, using extremely dense French knots to fill the flower centre. After about twenty minutes, she stopped and shook her hand. I asked what was wrong. She said: "My hand is sore. These knots take too much effort—I keep squeezing the needle tight." After a moment she added: "But it reminds me of that day in the field, bending down to pick vegetables—my lower back was sore in the same way."

She did not say "this reminds me of the pain"; she said "my lower back was sore in the same way"—for her, the two experiences were the same bodily feeling. Ingold says "making is knowing," and at that moment I truly understood it. Data embroidery is not using symbols to "represent" an experience external to it; it is re-activating that experience through the physical resistance of needle and thread passing through fabric. Her embroidered panel is not a record of pain but a re-enactment of pain. The density of stitches is not just data; it is the material trace of her fatigue from bending over in the vegetable field that afternoon. Previously, I had understood "data embroidery" as simply "translating data into embroidery"—still a binary way of thinking. Today I saw that embroidery itself is the process of data generation. It is not that data comes first and embroidery follows; rather, data is generated in the very act of embroidering.

DAY 7: THE MOMENT OF RECIPROCITY – AND UNFINISHED REFLECTION

On the final day, the embroidered panels were largely complete, and we sat together to share them. The stylistic variations were striking: some had full compositions with intense colours, others were sparse and airy with large empty spaces. Li Guilan looked around at the other women's work and said: "So everyone's 'dense' is different." There was no comparative judgement in her tone—it was an affirmation: we have all made our own experience clear in our own way.

At the end of the session, several women spontaneously asked: "Can we let people outside touch these embroideries and see where it hurts?" I sensed a subtle reversal—they had begun to view their embroideries as objects that could be "read" by others, and they wanted them to convey meaning beyond the community. From my initial aim of "getting them to express data through embroidery," to their own proposal of "letting the embroideries be read externally"—in that shift, a power relation had been reoriented.

But I must also record a sense of uncertainty. Throughout today, I kept thinking: after the project ends, I will leave with my photos and notes, while they remain in the village. This coding protocol, these embroideries, this narrative of "reciprocal design"—how much substantive change does it bring to their daily lives? Or does it primarily enrich my thesis? When Li Guilan said "let people outside see," she may have expected practical medical help or economic benefit, while what I can offer is probably just a prototype and a few papers. This gap unsettles me.

"Reciprocity" is not a state that can be "achieved"; it is an ongoing asymmetry that must be continuously confronted—what the researcher can give back is always less than what the community has given. Acknowledging this is more important than claiming "we have achieved reciprocity."

CONCLUDING REFLECTION

Over seven days in the field, I experienced a trajectory from "entering with preconceptions" to "being refuted by them," from "finding an interpretive framework" to "being corrected by the holder of that framework," and then to "reconceiving what knowledge production is." The most significant change was not an adjustment in research methods but a reconstruction of my self-understanding as a researcher—I am not someone who designs a better coding system, but someone learning how to become a person who allows local knowledge to correct my own presuppositions.

Li Guilan's phrase "you teachers really know how to think" will continue to remind me: the researcher's theoretical impulse is a privilege that requires continuous restraint. What I can do is not "speak clearly for them," but "prevent my theory from drowning out what they themselves say." This reflective journal itself is part of that restraint process—for every analytical sentence I write, I should ask myself: whose interpretation is this?

NO.2

1.Trifunovic, B.

Reshaping Cultural Preservation, Digital Innovation, and Technological Advancement

Trifunovic, B. (2025) Reshaping Cultural Preservation, Digital Innovation, and Technological Advancement PDT&C 54-4 Editorial. Preservation, Digital Technology & Culture, Vol. 54 (Issue 4),pp. 259-261. <https://doi.org/10.1515/pdtc-2025-0075>

This article has multiple inspirations for my topic: First, it emphasizes 'human-centered design' and 'accessibility,' indicating that the technological threshold must be lowered so that marginalized groups can truly become active participants rather than passive recipients; second, many cases in the article demonstrate how digital technology makes 'inaccessible' cultural heritage accessible, which is highly consistent with the logic of 'mutual rescue'; furthermore, the discussion of 'cultural design' as a 'value translation intermediary' provides a theoretical anchor for the core mechanism of 'reciprocal collaboration'; finally, the article's structural reflection on unequal resource distribution reminds us to avoid romanticizing 'reciprocity' and to confront inequalities in reality.

2.Xu, Q., Zhou, Z., Cao, N. and Liu, G.

From Struggle to Enjoyment: Investigating ICH Digital Inheritance Through the Lens of Vocational Education.

Xu, Q., Zhou, Z., Cao, N. and Liu, G., 2025. From Struggle to Enjoyment: Investigating ICH Digital Inheritance Through the Lens of Vocational Education. In: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25), April 26–May 1, 2025, Yokohama, Japan. New York: ACM, pp. 1–7. doi:10.1145/3706599.3719992.

This article takes embroidery inheritors as the research subjects, concretizing the concept of 'marginalized groups' as practitioners facing 'high learning barriers, rigid teaching models, social prejudice, and low economic returns,' which provides solid empirical material for arguing why 'marginalized groups need to be rescued.' Secondly, the issues revealed in the article, such as 'apprenticeships struggling to meet diversified needs' and 'traditional teaching lacking personalization,' ironically illustrate why 'reciprocal collaboration' is necessary—if cultural heritage clings to closed inheritance methods, it is difficult to attract new talent and cannot address the survival demands of modern practitioners. Furthermore, the 'intelligent tutor' framework proposed in the article (student model, teaching model, domain model) provides a feasible blueprint for 'empowering with digital technology,' especially the ideas of 'open-sourcing inheritors' knowledge into digital agents' and 'matching practical opportunities according to learners' levels,' which are concrete implementations of the 'mutual rescue' logic: technology both lowers entry barriers and provides inheritors' knowledge with new channels for dissemination and benefits. Finally, this article also reminds me that reciprocity cannot remain merely at the level of 'cultural identity'; it must respond to practical issues like 'economic returns' and 'professional dignity,' otherwise 'self-redemption' may still fail.

3.Liu, Z., Sun, H., Peng, Z., Duan, Z. and Pan, D.

Cultural transition by digital technologies: invasion or empowerment?

Liu, Z., Sun, H., Peng, Z., Duan, Z. and Pan, D., 2024. Cultural transition by digital technologies: invasion or empowerment? In: C. Gray, E. Ciliotta Chehade, P. Hekkert, L. Forlano, P. Ciuccarelli and P. Lloyd, eds., DRS2024: Boston, 23–28 June 2024, Boston, USA. London: Design Research Society. doi:10.21606/drs.2024.976.

This article, centered on the core question of 'invasion or empowerment,' directly responds to the presumption of 'mutual rescue'—the relationship between digital technology and marginalized communities as well as cultural heritage is not naturally reciprocal, but may bring unintended consequences such as 'cultural homogenization,' 'intensified individualization,' and 'commercial erosion.' The case study in the Eastern Tibetan region shows that while new technologies bring convenience to life, they also lead to the disruption of traditional ecological cycles, the disintegration of community collective rituals, and the commercialization of cultural symbols. When arguing for 'reciprocal collaboration,' one must face the power asymmetry and potential risks of technological intervention. The authors points out that marginalized groups are not passive recipients of technology, but demonstrate 'creative expression'—for instance, using billiard tables as platforms for carving Mani stones, or transforming fitness equipment into traditional saunas. These 'spontaneous design' practices precisely validate the proposition of 'marginalized groups as active agents.' The core of reciprocity should not be the unidirectional 'rescue' by external technology, but rather the discovery and empowerment of the existing creativity and adaptability of marginalized groups. The article's 34-day visual ethnography study, through coding of 'new objects, new materials, and object transformations,' reveals the micro-traces of technological infiltration. This detailed observation rooted in everyday material objects provides a practical research approach for studying 'reciprocity between marginalized groups and cultural heritage'—avoiding grand narratives and capturing the generative mechanisms of reciprocity from the changes of specific objects.

4.Tunstall, E.

Decolonizing Design: A Cultural Justice Guidebook

Tunstall, E., 2023. Decolonizing Design: A Cultural Justice Guidebook. Cambridge, MA: MIT Press.

This article has fundamental significance for critical reconstruction. Firstly, it directly points out that modernist design itself is part of a 'colonial project,' long embedding the exclusion and oppression of marginalized groups in Western-centric design theory and practice through the myth of 'technology improving life.' In this narrative, the implicit 'savior' is usually the Western designer or technologist, while the 'saved' are the marginalized communities—but this one-way logic is precisely what needs to be dismantled. This reminds you that before exploring 'reciprocal collaboration,' one must reflect on whether 'design' and 'digital technology' themselves carry colonial logic—if technological intervention is uncritically applied, so-called 'mutual salvation' —which still assumes that outsiders save insiders— may merely draw marginalized groups into a new process of cultural homogenization. Secondly, the 'decolonial design' proposed in the article provides you with a brand-new theoretical coordinate. The author advocates re-anchoring design theory and practice in the cultures and histories of global indigenous peoples, which exactly resonates with your core proposition of 'marginalized groups as active agents.' But even further, it requires you to ask: who is the subject of 'salvation'? Who has the right to define 'reciprocity'? In most collaborative projects, the external expert is still cast as the rescuer, and the local community as the recipient—yet this hierarchy contradicts the very idea of mutuality. If collaborative design is still led by external technical experts, the 'self-redemption' of marginalized groups is difficult to truly achieve. Therefore, the premise of reciprocity is the return of design power. Finally, the article emphasizes 'cultural justice' and 'repairing harm' rather than simple 'inclusion' and 'participation,' injecting ethical depth into your topic. It reminds you that the collaboration of digital technology and cultural heritage should not stop at functional optimization, but must confront historical injustice, placing the experiences and knowledge systems of marginalized groups at the core of the design process, making 'reciprocity' a practical path for repairing colonial wounds. True reciprocity demands that both parties save each other: marginalized groups save designers by grounding them in local knowledge, cultural values, and alternative epistemologies, while designers save marginalized groups by actively ceding power, resources, and authorship—so that salvation is no longer a one-way gift but a shared transformation.

ANNOTATED BIBLIOGRAPHY

5.Jovicic, S., Göbl, B., Verstappen, S. and Kayali, F.

Designing amidst uncertainty: drifting and byproducts in the intersection of anthropology and computer science

Jovicic, S., Göbl, B., Verstappen, S. and Kayali, F., 2025. Designing amidst uncertainty: drifting and byproducts in the intersection of anthropology and computer science. CoDesign, 21(2), pp.308–323. doi:10.1080/15710882.2024.2396945.

This article centers on the concepts of 'drifting' and 'by-products,' pointing out that in participatory design processes, genuinely valuable insights often do not come from predetermined research paths, but from participants' seemingly 'off-topic' creative expressions—such as a young person making an impressionist film about existential anxiety for a job-seeking app project, instead of directly discussing job hunting. This reminds us that when researching 'reciprocal collaboration' with marginalized groups and cultural heritage, one should not presuppose a linear script of 'technology saving culture' or 'culture empowering people.' Those seemingly 'irrelevant' narratives, emotions, and aesthetic expressions of marginalized groups may, in fact, be key entry points for understanding their real situations and subjectivities. At the same time, the article's emphasis on interdisciplinary equitable cooperation—anthropology and computer science 'are not each serving the other'—provides a reference for collaboration frameworks: digital technology should not be treated as a one-way tool, but as a medium for 'drifting' and co-generating meaning with marginalized groups. In other words, 'mutual rescue' is not a predetermined endpoint, but an open process full of uncertainty and by-products; researchers need to let go of their obsession with 'efficiency' and 'deliverables,' truly allowing the ways marginalized groups express themselves to define the pace and direction of collaboration.

6.West, P.

Designing in response to Indigenous sovereignties.

West, P., 2020. Designing in response to Indigenous sovereignties. In: Proceedings of the DRS 2020 International Conference: Tensions, Paradoxes, Plurality, 11–14 August 2020, Melbourne, Australia. London: Design Research Society. doi:10.21606/drs.2020.319.

This article reminds us that 'reciprocal collaboration,' if it does not first acknowledge the sovereign status of marginalized groups, may become a form of covert cultural colonization. The author points out that Western design knowledge systems assume themselves to be the 'master' or the 'neutral expert,' treating Indigenous peoples as the 'other' to be included or saved. This posture precisely replicates unequal power relations. Taking the Kulin people's 'Womin Djeka' (welcoming ceremony) as an example, true collaboration should be based on the 'visitor's' respect for the 'sovereign host': non-Indigenous designers must accept the limits of their own epistemology and acknowledge that Indigenous knowledge cannot be fully translated or absorbed into Western frameworks. For your topic, this means that digital technology empowerment should not presuppose a reciprocal exchange in which 'technology saves cultural heritage, and cultural heritage empowers marginalized groups.' Instead, we should first ask—who has the authority to define 'reciprocity'? Are marginalized groups (such as the ones you study) truly regarded as 'sovereigns,' rather than mere 'participants' or 'beneficiaries'? If the narrative of 'mutual salvation' strips away acknowledgment of sovereignty, land, and knowledge ownership, it may inadvertently re-objectify marginalized groups. Therefore, research needs to shift from 'collaborative design' to 'responding to sovereignty,' treating the epistemological boundaries set by marginalized groups as a prerequisite for technological intervention, rather than merely pursuing functional reciprocity.

7.Petkova, K.

Community cultural centers and the digital divide: between local engagement and systemic constraints

Petkova, K., 2025. Community cultural centers and the digital divide: between local engagement and systemic constraints. KIN Journal, 11(2), pp.1–14. doi:10.55630/KINJ.2025.110213.

This article provides important empirical support and critical reflection for my research topic. Using a community cultural center in Bulgaria as a case study, it demonstrates how institutions that serve both cultural heritage preservation and community service functions can spontaneously become key intermediaries for the elderly and vulnerable groups to bridge the 'digital divide,' even under conditions of extreme resource scarcity and generally low digitalization. This precisely responds to the core proposition of 'mutual rescue between marginalized groups and cultural heritage': while cultural centers help the elderly learn online payment and use electronic services, they also maintain their community vitality and cultural transmission through this process. However, the article also reveals the practical difficulties of this 'mutual benefit'—it relies heavily on individual enthusiasm and scattered projects, lacking systematic funding, policy, and training support, which results in most cultural centers remaining in a 'low-digital' state for a long time, or even struggling to sustain due to outdated equipment and unstable networks. This reminds us that the 'mutual rescue' enabled by digital technology should not be romanticized, and structural inequalities and uneven resource distribution must be confronted. At the same time, the article emphasizes 'local trust' and 'face-to-face, companion-style teaching,' highlighting the human-centered dimensions in collaborative design that technology cannot replace.

8.Lalopoulos, A.

Digital storytelling for intangible cultural heritage: a mobile application for the Museum of Asia Minor Culture

Lalopoulos, A., 2025. Digital storytelling for intangible cultural heritage: a mobile application for the Museum of Asia Minor Culture. In: Proceedings of the CAA 2025 International Conference.

This article uses the community museum of descendants of refugees from the 'Asia Minor Catastrophe' as a case study to demonstrate how digital technology (mobile applications, digital storytelling) can organically connect tangible exhibits with intangible cultural heritage (oral histories, everyday stories, traditional customs), thereby allowing the cultural memories of forcibly displaced 'marginalized groups' to be understood by younger generations and a global audience. This precisely echoes your logic of 'mutual salvation': cultural heritage is 'revitalized' through digital storytelling, and marginalized groups gain new channels for cultural expression and identity recognition through technology. However, this study also brings necessary reflections — although the applications enhance visitor engagement, the design and development are still led by external researchers, with community members participating more as information providers rather than decision-makers. This reminds you that 'reciprocal collaboration' cannot stop at 'cooperation' in technological presentation but requires questioning whether design power is truly returned to marginalized groups; otherwise, 'self-salvation' may still degenerate into a one-way narrative empowered by external technology. Moreover, the practical difficulties revealed in the article, such as resource scarcity in small museums and limited digital content, also warn that research must acknowledge structural constraints and avoid romantically imagining 'reciprocity'.

9.Giglitto, D., Ciolfi, L., Lockley, E. and Kaldeli, E., eds.

Digital Approaches to Inclusion and Participation in Cultural Heritage: Insights from Research and Practice in Europe

Giglitto, D., Ciolfi, L., Lockley, E. and Kaldeli, E., eds., 2023. Digital Approaches to Inclusion and Participation in Cultural Heritage: Insights from Research and Practice in Europe. Abingdon: Routledge. doi:10.4324/9781003277606.

This introduction provides an important theoretical framework and empirical reference for my research topic. The book clearly centers on the question of 'how digital technology can promote the participation of marginalized groups in cultural heritage,' emphasizing that cultural participation is a fundamental human right, and that means such as digital storytelling and community co-creation can effectively foster social inclusion. Among these initiatives, the MEMEX project enables marginalized groups to reshape cultural expression through AR stories; the CultureLabs platform lowers collaboration barriers through a 'recipe' approach; and decolonization practices targeting Sámi visual archives directly embody the logic of 'reciprocal collaboration'—cultural heritage gains diverse interpretations through the involvement of marginalized groups, who in turn use digital tools to articulate their culture and reconstruct their identities. However, the book also raises critical reflections: value conflicts among stakeholders, power imbalances between institutions and communities, and biases embedded in technological design all risk making 'reciprocity' merely superficial.

10.Bavi, A.

Seeing Through Touch: Creating Inclusive Experiences for Blind and Partially Sighted Museum Visitors Through 3D Printing

Bavi, A., 2025. Seeing Through Touch: Creating Inclusive Experiences for Blind and Partially Sighted Museum Visitors Through 3D Printing. PhD thesis. University of British Columbia (Okanagan).

This doctoral dissertation provides significant empirical insights and critical reflective value regarding the topic. It centers on the marginalized group of blind and partially sighted individuals (BPS) and demonstrates how 3D printing technology can transform Inuit cultural heritage (such as the 'Owl' sculpture) into tactile models, thereby creating a reciprocal relationship between the 'accessibility' of cultural heritage and the 'participation' of marginalized groups: cultural heritage is revitalized through digital restoration and tactile reproduction, while the BPS community gains cultural identity recognition and aesthetic experience through touch. More importantly, the dissertation deeply examines the ethics of collaboration with the Winnipeg Art Gallery, the OCAP® principles (Ownership, Control, Access, Possession), and intellectual property agreements, revealing that 'reciprocity' is not merely a one-way technical empowerment but must be grounded in power sharing, community sovereignty, and transparent agreements.

11.Ingold, T.

Making: Anthropology, Archaeology, Art and Architecture.

Ingold, T. (2013) Making: Anthropology, Archaeology, Art and Architecture. London: Routledge.

Ingold's (2013) core insight on "knowing from the inside" directly challenges the conventional technological trajectory in digital empowerment of cultural heritage—a model in which external experts define coding standards while marginalised groups serve merely as "information providers" or "cultural executors." He argues unequivocally that genuine knowledge is not "information" acquired from the outside, but a capacity that grows through sustained embodied dialogue with materials, tools and the world. This judgment compels us to transform the passive narrative of "being saved" into the agentive process of "self-redemption"—the starting point of digital empowerment should not be the "deployment" of technological solutions, but rather the correction of technological frameworks by the marginalised groups' craft epistemologies. Specifically, when the embroiderers refused to standardise stitch density and instead reconstructed coding rules through the embodied metaphor "dense stitches mean long-lasting pain," they precisely enacted what Ingold calls "following"—researchers abandoning their external posture as planners and learning to "correspond" with materials, movements and bodily rhythms. Consequently, the core of a "reciprocal design collaboration model" lies not in how technological systems can more efficiently represent local knowledge, but in how they can be re-architected to learn those embodied differences that cannot be captured by standardised metrics—this is precisely the epistemological shift from "being saved" to "self-redemption."

No.	Author(s) (Year)	Key Points / Main Content	Relevance to "From Being Saved to Self- redemption / Reciprocal Collaboration"
1	Trifunovic (2025)	Human- centred design, accessibility, cultural design as value translation intermediary	Provides theoretical anchor: reciprocal mechanism needs to lower technological thresholds and avoid romanticising "reciprocity"
2	Xu et al. (2025)	Embroidery inheritors face high learning barriers, low returns; intelligent tutor framework	Empirical evidence why "marginalised groups need to be rescued"; concrete paths for technological empowerment (open- sourcing knowledge, matching practical opportunities)
3	Liu et al. (2024)	"Invasion or empowerment?" Case from eastern Tibet shows cultural homogenisation, commercialisation risks	Warns about power asymmetry of technological intervention; emphasises existing "spontaneous design" of marginalised groups as core of mutual rescue
4	Tunstall (2023)	Decolonising design; modernist design as part of colonial project	Critical foundation: the premise of mutual rescue is returning design power; must confront historical injustice
5	Jovicic et al. (2025)	"Drifting" and "by- products"; equitable interdisciplinary cooperation	Mutual rescue is not a linear script but an open process; marginalised groups' "off- topic" expressions are key entry points
6	West (2020)	Responding to Indigenous sovereignties; collaboration must be based on 'visitor vs. sovereign host"	If mutual rescue narrative strips sovereignty, it may re- objectify marginalised groups; need to shift from 'co- design' to "responding to sovereignty"
7	Petkova (2025)	Bulgarian community cultural centre helps elderly bridge digital divide	Real- world case of mutual rescue, while exposing structural resource inequality; emphasises local trust and companion- style teaching
8	Lalopoulos (2025)	Digital storytelling at museum of Asia Minor refugee descendants	Embodies mutual rescue logic, but shows design power still lies with external researchers – self- redemption not truly achieved
9	Giglitto et al. (2023)	Digital approaches for inclusion of marginalised groups in cultural heritage (MEMEX, CultureLabs, etc.)	Provides multiple case supports for mutual rescue, while raising critiques such as value conflicts and institutional power imbalances
10	Bavi (2025)	Blind and partially sighted people engage with Inuit cultural heritage via 3D- printed tactile models	Deeply demonstrates that mutual rescue must be grounded in power sharing, community sovereignty, and transparent agreements (OCAP principles)

Table 1: Core Information of the Literature (Theme, Author, Key Points, Relevance to Research)

Theory / Concept	Source	Core Meaning	Use for Research
Human- centred design / Accessibility	Trifunovic (2025)	Technological thresholds must be lowered so that marginalised groups become active participants	As a design principle for reciprocal collaboration
Intelligent tutor framework (student- teaching- domain model)	Xu et al. (2025)	Open- source inheritors' knowledge into digital agents, match practical opportunities according to learners' levels	Concrete technical implementation path: how mutual rescue can be realised
Invasion vs. empowerment	Liu et al. (2024)	Technology can bring cultural homogenisation, individualisation, commercial erosion	For critiquing the assumption that "technology is inherently good"; introducing power asymmetry analysis
Decolonising design / Cultural justice	Tunstall (2023)	Design is not neutral; it must repair harm and return design power	As the fundamental stance of the research; precondition for shifting from "being saved" to "self- redemption"
Drifting & by- products	Jovicic et al. (2025)	Truly valuable insights come from participants' seemingly off- topic expressions	Methodological guidance: do not presuppose a linear mutual rescue script; keep the process open
Responding to sovereignty (not just co- design)	West (2020)	Non- Indigenous designers must accept their epistemological limits and respect sovereign boundaries	Redefine "reciprocity"; not as functional exchange but as sovereignty recognition
Local trust / Companion- style teaching	Petkova (2025)	Technology cannot replace face- to- face, low- threshold, companion- based learning	Human dimension in mutual rescue design
OCAP® principles	Bavi (2025)	Ownership, Control, Access, Possession	Ethical and legal framework for reciprocal collaboration

Table 2: Key Theoretical Concepts and Frameworks

Case / Method	Source	Key Practices	Key Practices
Vocational education for embroidery inheritors	Xu et al. (2025)	Intelligent tutor, open-sourcing knowledge, matching practice by level	Concrete interaction model design for technological empowerment
34-day visual ethnography in eastern Tibet	Liu et al. (2024)	Coding "new objects – new materials – object transformations" to capture micro-traces of technological infiltration	Avoid grand narratives; capture reciprocal generation mechanisms from changes in everyday objects
Bulgarian community cultural centre	Petkova (2025)	Low-tech, companion-based digital literacy under extreme resource scarcity	Real-world reference for low-cost mutual rescue; emphasises trust and sustainability
Asia Minor refugee museum app	Lalopoulos (2025)	Digital storytelling linking tangible exhibits and oral histories	Reminder: if design power is not truly returned, it remains an external technical narrative
MEMEX project / CultureLabs	Giglitto et al. (2023)	AR stories, community co-creation, "recipe" approach to lower collaboration barriers	Reusable models for facilitating marginalised group participation
3D-printed tactile models + OCAP	Bavi (2025)	Blind community experiences Inuit heritage through touch, with IP agreements	Necessity of power sharing and legal protocols in mutual rescue

Table 3: Methodologies and Case Studies

NO.3



knowing from the inside
knowledge grows

embodied practice

Tim Ingold's opening proposition in *Making*—that of “knowing from the inside” —first and foremost implies that knowledge is not passively received “information” but rather a capacity that grows through sustained embodied practice. Drawing on his Saami mentors’ injunction to “know for yourself,” Ingold argues that genuine understanding cannot be transmitted through external instruction—instruction can only offer superficial directives, never replacing the learner’s ongoing dialogue with materials and with the world in action. Knowledge, therefore, is not a representation of the world; it is a mode of being: we are able to know the world not because we stand outside it as observers, but because we are already of the world, co-becoming with it through participation and response.

On this basis, Ingold further distinguishes between anthropology and ethnography as radically different enterprises. Ethnography aims at descriptive documentation about the world, reducing participatory experience into analysable “data”; anthropology, by contrast, should be a transformative practice with the world, in which the knower is changed through action. He condemns the former as an epistemological regression—it degrades living perceptual practice into qualitative material and severs “knowing” from “being.” In opposition, “knowing from the inside” demands that the researcher abandon the external posture of the spectator and immerse herself in the immanent flow of co-existence with all things. This flow is not an isolated intellectual activity, but is woven through rhythmic correspondences among hands, eyes, perception and materials.

representation

mode of being
co-becoming

This insight resonates deeply with the co-design practice in my paper. When the embroiderers, during the negotiation of coding rules, refused to standardise stitch density—Sister Pan’s assertion that “everyone’s stitches are different in tightness”—this is precisely a practical expression of “knowing from the inside.” Their knowledge is not a set of “information” that can be extracted from their bodies and hands and calibrated by external metrics; it is an embodied baseline formed through years of repeated dialogue with threads and fabric. Within their epistemological framework, the tightness or looseness of stitches is not a “individual difference” to be eliminated, but an irreducible signature of personal bodily narrative. When the ontological metaphor “dense stitches mean long-lasting pain” spontaneously emerged from bodily experience, and when this led us toward personal calibration rather than uniform standards, we effectively underwent a shift—from an external, standardising stance to a co-becoming with the embroiderers’ craft knowledge.

However, the most sobering lesson for me was Sister Pan’s polite correction of our over-theorisation—“You teachers really overthink it; I didn’t think that much” —which precisely echoes Ingold’s warning: researchers, too, can, in the name of “understanding,” translate participants’ improvisational expressions into academic language and then appropriate them. Genuine epistemological humility is not a one-time reflective epiphany, but a practice of sustained vigilance throughout the entire process of research and writing. The most important lesson I have drawn from Ingold is this: the “co-” in co-design does not mean that designers and participants sit at the same table and draw the same picture; rather, it requires the researcher to abandon the external stance of the “planner” and learn how to follow the lines and rhythms in the embroiderers’ hands, allowing knowledge to grow organically from the interweaving of materials, movements and perception. The design of the personal calibration system is but an institutionalised expression of this “following” —the technological system does not seek to unify differences, but to learn from them, which is precisely a correction of technological rationality’s standardising presupposition by “knowing from the inside.”

LITERATURE ASSIGNMENT

NO.4

Research Methods in Practice
Researcher: Yuki
Project Affiliation: Reciprocal Design Research on Marginalised Communities and Cultural Heritage – Co-designing Embroidered Body Data with Rural Women in Lanzhou
Research Method: Semi-structured Interviews + Supplementary Survey Questionnaire
Specific Aspect Investigated: Embroiderers' Attitudes towards "Standardisation of Stitch Density" and their Acceptance of Alternative Coding Schemes

研究方法的实践
研究者：雨琦
项目关联：边缘群体与文化遗产的互惠设计研究——兰州农村女性身体数据刺绣共同设计
研究方法：半结构化访谈 + 补充性问卷调查
研究具体方面：绣娘对“针法密度标准化”的态度及其替代性编码方案的接受度

A. SURVEY QUESTIONNAIRE

Survey Participants: 12 rural women from Yuzhong County, Lanzhou, participating in the co-design workshop
Survey Period: 10–11 June 2026
Administration Method: Oral administration – questions read aloud one by one, with the researcher ticking responses on behalf of participants (approximately half of the participants were not literate)
Valid Responses: 12 (100% response rate)

Item	Question	Options	Responses
1	Do you think that "pain duration" can be distinguished using a unified three-level stitch-density standard (sparse / medium / dense)?	A. Yes, completely	0
		B. Roughly, but with adjustments needed	2
		C. No, because everyone's hand strength is different	10
2	If you were asked to personally define your own "sparse," "medium," and "dense," would you find it difficult?	A. Not difficult – I've been embroidering for decades, I know my hand	9
		B. A little difficult, but manageable with a few tries	3
		C. Very difficult – I wouldn't know what the standard should be	0
3	Do you think that mild pain is suitable to be represented by cool colours (e.g. light blue, light green)?	A. Yes – coolness feels like mild pain	4 (all under 40)
		B. No – pain should be warm or red tones	7 (all over 50)
		C. No preference	1
4	If the final embroidered panels show individual style differences (e.g. some stitch densely, some sparsely), do you consider this:	A. A problem – they should be unified	0
		B. Not a problem – each person embroiders in her own way	12
5	Would you want your embroidered panel to be seen and interpreted by outsiders?	A. Yes	10
		B. No	0
		C. Hadn't thought about it	2

Survey Questions and Results

Brief Survey Analysis:

The results of Question 1 directly confirm the "standardisation clash" I experienced on the first day of fieldwork – only 2 women considered a unified standard roughly acceptable, while 10 explicitly rejected it, all citing "different hand strength."
Question 3 reveals a clear generational divide: all those who chose "cool colours are suitable for mild pain" were under 40, while all opponents were over 50 – consistent with the generational colour disagreement recorded in Day 5 of my reflective journal.
The result of Question 4 is particularly noteworthy: all 12 participants affirmed that individual style differences are "not a problem." This suggests that their insistence on manual variation is not an isolated opinion but a shared value judgement within the community.
Question 5 shows that 10 participants hope their work will be seen by outsiders, indicating that the project's external dissemination goal is not solely driven by the researcher but aligns with community aspirations.

B. INTERVIEW TRANSCRIPTS

Interview Format: Semi-structured, one-on-one interviews conducted during breaks in embroidering (Day 6), held in the village activity room and in courtyards. The interviews were conducted in the local Lanzhou dialect, with second researcher Li Minghui providing interpretation into Mandarin. The following are selected excerpts of key exchanges.

Interviewee 1: Li Guilan (age 54, 38 years of embroidery experience)

Interview Date: 14 June 2026, morning

Location: By the window of the village activity room

Core Theme: Rejection of standardisation and the feasibility of personal calibration

Q: Yesterday you said, "everyone's sparse and dense are different." So if we don't impose a unified standard and let each person use her own definition of sparse and dense, do you think that would work?
A: Yes, it would. I know what my "dense" looks like – I've been embroidering for decades, my hand knows.
Q: But your "dense" might not be the same as Wang Xia's "dense." Would outsiders looking at it be able to understand?
A: (pauses) They can take their time figuring it out. She embroiders her way, I embroider mine. Each person's pain is her own. If you force me to embroider exactly like her, then what I stitch won't be my lower back anymore.
Q: If we built a system that could learn your stitch pattern – no matter how you embroider, the machine would recognise it – would you like that?
A: That would be fine. If the machine recognises mine, that's okay – as long as I don't have to copy someone else's style.
Q: Is there anything about this approach that you think might not work well?
A: (laughs) I'm just worried that once you leave, the machine will have no one to look after it, and it might break.

RESEARCH METHODS IN PRACTICE

Interviewee 2: Wang Xia (age 32, 12 years of embroidery experience, still learning)

Interview Date: 14 June 2026, afternoon

Location: Wang Xia's home courtyard

Core Theme: Identity anxiety among younger embroiderers and technological sustainability

Q: You have fewer years of embroidery experience than others. Did you find it harder to produce your personal calibration samples?

A: Yes, a bit. Aunt Li and the older women have embroidered all their lives – their hands just know where the needle should go. I still have to think about it sometimes, and if it doesn't look right, I undo it and re-stitch.

Q: If your calibration samples turn out less accurate than theirs, and your embroidered panel doesn't get recognised as well in the system, would you feel uncomfortable?

A: (silence) ...Then don't tell me about it. (laughs) Actually, I know I'm not as skilled as them, but I still want to participate.

Q: Is there anything that would make it easier for you?

A: If I could watch them do it first, and then do mine, that might help. Or you could let me do it a few times – the first time doesn't count, I can practise before I do the official one.

Interviewee 3: Grandma Liu (age 67, over 50 years of embroidery experience)

Interview Date: 14 June 2026, afternoon

Location: Steps outside the village activity room

Core Theme: Deep-seated reasons behind the generational colour disagreement

Q: Last time you said, "blue is not pain." But Wang Xia and others think light blue can represent mild pain. Why do you think there's this difference?

A: She's young – she hasn't suffered the way I did. When I was young, working in the fields under the blazing sun, heavy labour – when my back hurt, it was a burning pain, like fire. That cool, breezy feeling – that's wind, not pain.

Q: If we asked you to use blue to express something you feel, could you?

A: That would be coldness. In winter, when my knees get cold and ache – that's a cold ache, different from burning pain. If you insist that I use colours to distinguish them, I'd have to separate them – one is the burning kind, the other is the cold kind.

Q: Could we keep both options available for you when we design the colour coding?

A: That's fine – as long as I don't have to use blue to represent mild pain.



Brief Survey Analysis:

My research is cross-cultural in two intertwined senses: first, there is a cultural and educational gap between myself as the researcher—trained in Western academic paradigms—and the embroiderers, who are rural Chinese women embedded in local craft traditions and bodily practices; second, the very object of inquiry—the encounter between traditional hand-embroidery and modern digital coding systems—constitutes a cultural interface where different knowledge systems and value orientations meet. In this sense, the cross-cultural nature is both interpersonal and epistemological. In this methods exercise, I employed a combination of surveys and semi-structured interviews, focusing on "embroiderers' attitudes towards standardisation of stitch density and their acceptance of alternative coding schemes" as the specific investigative lens. The most important lesson this exercise taught me is: the validity of survey questionnaires in cross-cultural research is far more fragile than I had assumed, and interviews are truly the method that can reach the epistemological level.

After collecting the survey responses, I found that Question 1 – "Can a unified standard for sparse/medium/dense be used?" – received a clear "no" from 10 out of 12 participants. On the surface, the data appeared neat. But this neatness obscured a critical fact: the reasons for their rejection varied. Li Guilan's rejection was based on "different hand strength"; Wang Xia's was rooted in fear of social comparison; and Grandma Liu's came from "a lifetime of habit that can't be changed." Behind the same "no" tick-box lay different bodily histories and social positions. Had I relied solely on the questionnaire, I would have obtained a clean statistic – but I would have understood nothing about the "why."

The interviews revealed what the questionnaire entirely missed: the unnegotiable firmness in Li Guilan's tone when she said "each person's pain is her own" – this was not an "opinion" about the proposal, but a declaration of bodily sovereignty. Her "no" was not a starting point for negotiation; it was a principle. On a questionnaire, that "no" would have been just a checked box. In the interview, it was a complete, embodied knowledge position.

Another critical realisation came from the tension between the survey data on Question 5 and the subsequent interview responses. The survey showed that 10 participants "hoped outsiders would see their embroideries," which led me to optimistically assume that the project's external dissemination goal was closely aligned with community aspirations. But when I probed further in the interviews – "who exactly do you want to see them, and for what purpose?" – the answers became vague: "outsiders," "let more people know." Their imagined "outside" did not neatly overlap with my understanding of "academic dissemination" or "public exhibition." The survey made me think we had a consensus; the interviews revealed that consensus was hollow.

Based on this methodological exercise, I have adjusted my approach for the remainder of the study: (1) from now on, I will only use questionnaires for collecting demographic background information, not for measuring attitudes or opinions; (2) all data on "acceptance," "consensus," and "evaluation" must be sourced primarily from open-ended interviews; (3) prior to formal coding, I will add a round of group discussion based on interview findings to test individual views against the collective public sphere.

Perhaps the deepest takeaway is this: a questionnaire asks "what do you choose?", whereas an interview asks "what do you think?" When it comes to research that involves bodily experience and cultural knowledge, "what do you think" matters far more than "what do you choose" – because the latter can be standardised, while the former is precisely what epistemological justice must respond to.

NO.5

Assignment 5: Artefact & Reflection

Researcher: Yuki

Title of Artefact: Stitches of Pain: An Interactive Digital Archive of Embodied Narratives by Rural Women in Lanzhou

Type: Interactive Digital Narrative Web Prototype

Presentation Format: Prototype demonstration video + key interface design mock-ups (attached after this reflection)

作业 5: 作品与反思 (Artefact & Reflection)

研究者: 雨琦

作品名称: 《疼痛的针脚: 兰州农村女性身体叙事交互档案》

作品类型: 交互式数字叙事网页原型 (Interactive Digital Narrative Web Prototype)

作品呈现形式: 网页原型演示视频 + 关键界面设计稿 (附于本页后)

DESCRIPTION OF THE ARTEFACT

This work is an interactive digital archive prototype centred on embodied pain narratives. Its primary material is drawn from the personal embroidered pain panels created by twelve rural women in Yuzhong County, Lanzhou, along with the oral accounts recorded during their creative process.

My initial plan for this project was to develop a researcher-led coding framework that would systematically translate the embroiderers' pain diaries into quantifiable stitch patterns, with predefined categories and density metrics that I assumed would be uniformly applied across all participants. However, the form of this artefact was not determined unilaterally by the researcher. Rather, it emerged from the collective request of the embroiderers themselves on the seventh day of fieldwork. After completing their embroidered panels, Li Guilan and nine other women expressed in the survey their wish that "outsiders could understand these bodily stories." Wang Xia specifically proposed: "If you could touch the embroidery and see where it hurts, that would be good." This request became the initial impetus for the work. The research team recognised that if reciprocal design ended with the completion of the embroideries, the bodily experiences stitched into them would remain confined within the village and would not respond to the women's desire to be seen by outsiders.

The final form of the work is a low-tech, touch-friendly web page composed of three core layers:

Layer 1 (Surface) – High-resolution digital images of each embroidered panel, with the floral motifs (representing different body parts) set as clickable "hotspots."

Layer 2 (Interaction) – When a user clicks on a floral motif, the page triggers two actions: (a) a gradual animation appears on the corresponding body part, visually representing pain through changing colours and textures, with rhythm and intensity defined according to the maker's personal calibration card; and (b) the maker's own oral narrative is played automatically, in dialect, describing the specific sensation and daily context of that pain.

Layer 3 (Depth) – Each embroidered panel is accompanied by a digitised display of the maker's personal calibration card, showing her hand-stitched "sparse-medium-dense" density samples and her custom colour chart. This makes explicit to the viewer that each panel's coding rules are personal and that no cross-individual unified standard exists.

At present, the prototype exists as a high-fidelity static demonstration version (with complete interface design and simulated interaction screen recording). The core technical functions—image recognition and real-time audio synchronisation—remain to be fully developed. However, its design logic already fully articulates the outcomes of the co-design process and its epistemological position.

ARTEFACT & REFLECTION

SIX ACTIONS FOR COLLECTING INFORMATION ABOUT THE ARTEFACT AND ITS CREATIVE PROCESS

Drawing on Ingold's (2013) concept of "knowing from the inside" in Making—that knowledge and form are not fully preconceived before action and then mechanically executed, but rather "grow" through the maker's ongoing dialogue, resistance, and response with materials—I have designed the following six actions to collect critical information about this digital work and its creation. The core concern underlying these actions is: when craft experience is translated into digital media, what is preserved, what is distorted, and what is lost? And what do these losses themselves reveal?

Action 1: Reverse-validation interviews with embroiderers about their digitised panels

Each panel maker will be invited to view how her embroidery is presented on the webpage—the animation speed, the digital colour reproduction, the length and sequencing of the audio clip. I will ask each one: "Does the pain in this animation feel like the pain you felt while embroidering it?" I will pay particular attention to and record moments when she says "not quite" or "a bit different", and probe further: "What is different?" These "not quite" moments are precisely where the digital medium resists the craft experience. Ingold argues that knowledge emerges from the interaction between maker and material—and when the material (here, the digital interface) refuses to comply with the logic of craft experience, that zone of resistance becomes the most intensive site of dialogue between two epistemologies. If Li Guilan says, "this pain moves too fast—my pain lingers and won't go away," then Action 1 will capture a bodily temporality that a technical parameter (animation speed) cannot reproduce.

Action 2: Gesture and gaze tracking during interface selection

When the embroiderers first browse the web prototype, I will give no verbal guidance and will only observe and record their operational behaviour: where do their fingers first touch? On which motif do they linger longest? Do they click repeatedly on the same spot? Do they notice the "personal calibration card" entry in the corner? Do they proactively click on the maker's name? These behavioural data will be manually recorded as a "heatmap of interest." The Ingoldian logic of this action lies in treating the viewing of a digital interface not as a purely visual activity, but as an embodied activity—how the hand touches the screen and how the gaze moves between image and text are themselves ways of negotiating with digital material "from the inside." The trajectories of these behaviours will reveal how the embroiderers perceive the translated form of their own work: which translations feel foreign (and are thus avoided), and which feel familiar (and are thus revisited).

Action 3: Automated interaction logging

During the public testing phase (with 20 external visitors invited to participate—including county cultural centre staff, university students from Lanzhou, and general urban residents), the system will automatically record the following data: total clicks per embroidered panel, individual click distribution for each floral motif (body part), dwell time per click, whether the full oral audio was played, and whether the personal calibration card page was viewed. These log data will answer a question that interviews alone cannot address: which bodily narratives are more easily "read" by external viewers (high clicks, long dwell time), and which are systematically overlooked? The latter may point to aspects of the coding protocol that are culturally less legible—for instance, if embroideries by older women receive significantly fewer clicks than those by younger women, this may indicate that the "burning pain" expressed in warm colours is less resonant for external viewers, or that external viewers unconsciously overlook older women's bodily narratives.

Action 4: Researcher's own "translation decision log"

The webpage production process involves a series of subjective judgments that cannot be covered by technical manuals: translating hand-dyed silk colours into screen colour codes (e.g. "does this blue lean grey or green?"; "should the saturation of this red be preserved or reduced to fit the display gamut?"); and editing dialect audio recordings into playable narrative segments (e.g. "is this pause a breath or hesitation? Should it be cut?"). For each such decision, I will create an independent record: writing down the reasoning, hesitations, and final choice at the time. The purpose of this "translation decision log" is not to prove the "accuracy" of the translation—because no accurate translation from craft to digital media is possible—but to reveal a fact typically erased by technical documentation: so-called digitisation consists of a series of embodied, non-replicable judgments. This log itself is a response to Ingold: if we acknowledge that knowledge is generated through dialogue with materials, then the researcher's own dialogue with digital materials should itself be recorded as data.

Action 5: Public feedback recording during collective viewing

After the web prototype is completed, I will organise a collective viewing session. The embroiderers will see their panels on screen for the first time, being clicked and commented on by "outsiders." I will record the entire session on video, but my primary focus will be on their reactions when watching others being clicked—is it pride, tension, discomfort, or indifference? When one woman's pain story is repeatedly clicked while another's is barely touched, the exchange of glances and whispered conversations among the onlookers will reveal what the shift from "private bodily narrative" to "public cultural exhibit" means for the participants. Ingold's "from the inside" here extends to "opening from the inside outward"—this opening is not a loss of internal knowledge but another form of knowledge production: it allows the embroiderers, for the first time, to observe from an external perspective how their own experiences are received.

Action 6: Dedicated archiving of recognition error cases

The technical system may show recognition errors for some embroiderers' stitch densities (especially older women with looser hand strength). In engineering logic, these are "noise to be optimised." However, I will not treat them as errors. Instead, I will create a separate folder named "Pains the System Has Not Yet Learned to Read." I will collect these "failure" cases and interview the makers themselves: "The machine can't read your stitches—what do you think is the reason?" If Grandma Liu's "dense" panel is recognised by the system as "medium," it may be because her French knots, after fifty years of changing hand strength, have morphological features that can no longer be captured by a recognition algorithm based on uniform stitch-spacing assumptions. This "failure" precisely defines the real boundary of the personal calibration system—and also defines which components of craft knowledge the technical system cannot (and perhaps should not) attempt to capture. As Ingold emphasises, the resistance of materials (here, needle and thread) is not an obstacle to making but a necessary condition for the generation of form.

COMPREHENSIVE REFLECTION THROUGH THE LENS OF INGOLD'S "KNOWING FROM THE INSIDE"

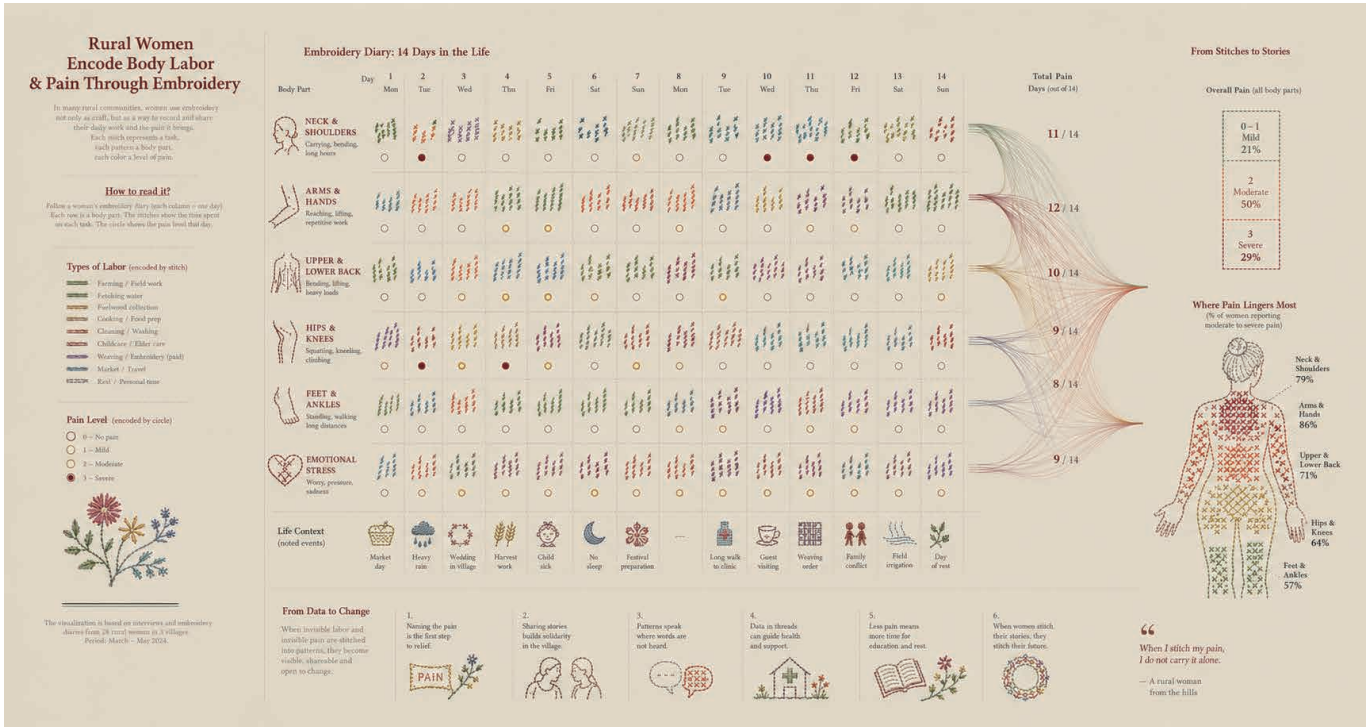
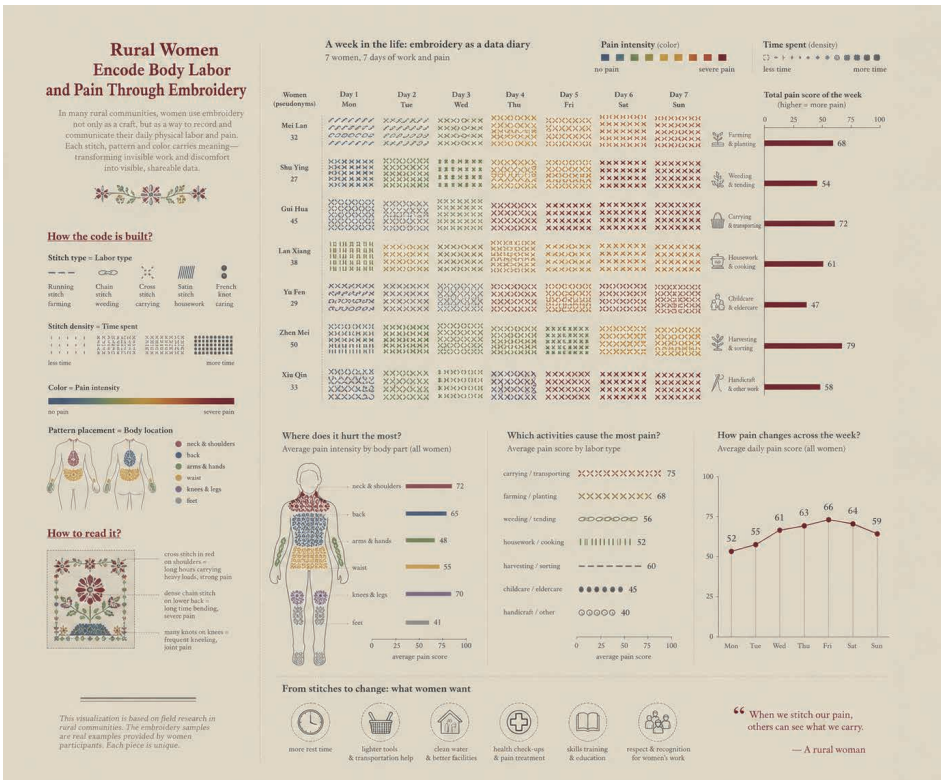
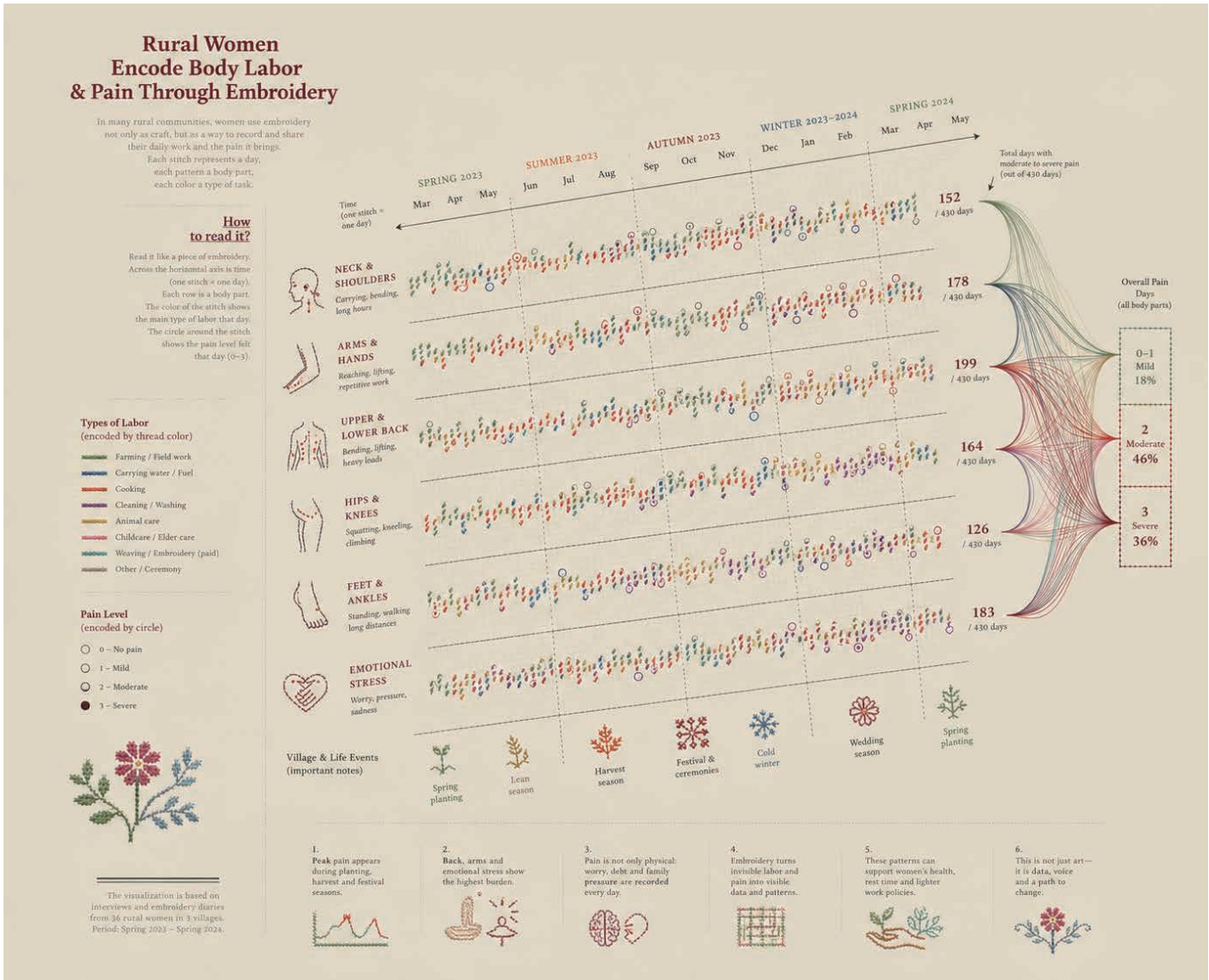
The common logic underlying the six actions above is to treat the web prototype as an ongoing epistemological event, rather than a "finished" and "delivered" product. Ingold's central argument in Making is that making is not a process of imposing a pre-existing design upon passive materials, but of allowing form to "grow" through continuous dialogue with materials. In conceiving and initially developing this work, I experienced three re-understandings of this argument.

The first understanding came when digitising the colours of the embroidered panels. Grandma Liu's panel used deep red to express "burning pain." When I attempted to reproduce this colour on screen, I found a significant gap between the colour captured by the scanner and what I saw with my eyes. I spent hours fine-tuning the colour temperature in pursuit of an "accurate reproduction." But when I paused to ask myself "what does accuracy mean here?", I realised that the scanner records reflected light data, whereas Grandma Liu chose her colour with reference to the burning memory of her body labouring in the fields under the sun when she was young—there is no common metric between these two. In the end, I added a note on the webpage: "Screen colours are digital approximations only; the colour experience of the physical embroidery must be encountered in person." This seemingly technical note is, in fact, an epistemological position: acknowledging that some knowledge refuses translation, and acknowledging the boundaries of digital media, is itself a practice of Ingoldian humility.

The second understanding came when adjusting the animation speed. Initially, I designed a smooth, rapid animation (0.5-second completion) to create an immersive "touch-to-feel" effect. However, during an informal preview, Li Guilan remarked: "It moves too fast—it doesn't feel like pain. Pain is slow and dull; it lingers." I subsequently changed the animation duration from 0.5 seconds to a slow 2.5-second ease-out. This technical adjustment was, at its core, a letting-go of the default aesthetics of digital media (fluidity, instantaneity) in favour of the temporality of bodily experience (slowness, viscosity). For Ingold, this is not "optimisation"; it is allowing the grammar of the digital material to be corrected by craft knowledge.

The third understanding—and the most personally corrective—came when designing the collective viewing session. I realised that the most essential function of this webpage is not to "display data" but to allow the embroiderers to see how others view their bodily narratives. The embroidered panels move from private, domestic objects into a public interface that can be arbitrarily clicked by external viewers—this displacement is not a neutral technical operation but a reconfiguration of social relations. Ingold emphasises that making is a "dialogue with materials," but through this work I have added a dimension that Ingold relatively neglects: the dialogue is not only between maker and material, but also between maker and future viewers—even if those viewers are not yet present. The embroiderers' request to "let outsiders understand" is itself an imagination of publicity that originates "from the inside." What this web prototype is meant to carry is precisely the material trace of that imagination.

Final conclusion: For me, "knowing from the inside" is not a theoretical framework to be "applied" to the work; it is a question I must repeatedly ask myself at every technical decision point—when choosing a visual style, when designing an interaction flow, when selecting start and end points for an audio clip—I must ask: whose logic is driving this? Is the embroiderer's bodily rhythm being heard? If this digital work has any value, it lies not in its "accurate" representation of pain data, but in its preservation of what cannot be precisely represented—and in its honest marking of those limits within the interface itself.





兰州大学威尔士学院
LANZHOU UNIVERSITY WALES COLLEGE

Invisible Pain and Visible Stitches

—Why Use Embroidery to Record the Body?

Stitching the Invisible · 织隐

Research Team

Lanzhou University Wales College
Yuki



Scan for full paper

B ■ Invisible Bodies: Rural Women's "Data Absence"

"My back hurts so much I can't straighten up—but nobody can see it."
—A tea-picking woman in Embroidery Village, Guizhou

Transplanting rice, picking tea, weaving, carrying heavy baskets... day after day, physically demanding labour stitches fatigue and pain deep into the bodies of rural women. Yet these bodily experiences are rarely recorded:

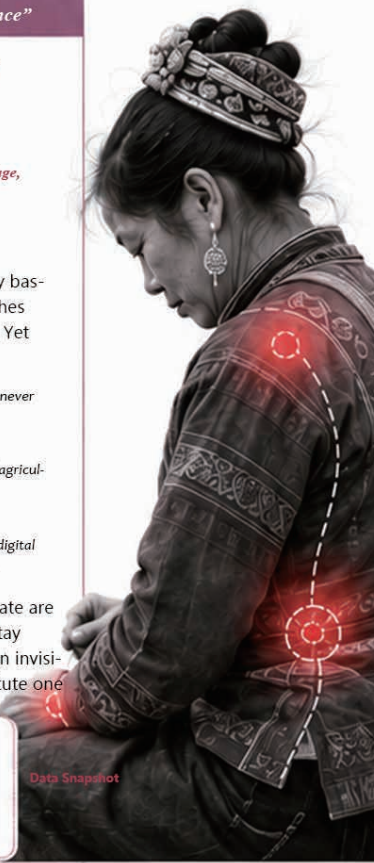
They do not appear in hospital records (many women have never received a formal pain diagnosis).

They are absent from government statistics (domestic and agricultural work is not covered by occupational injury data).

They are invisible in the data streams of wearable devices (digital health tools remain detached from their everyday worlds).

When the power to collect data and the right to narrate are monopolised by external systems, women's bodies stay trapped in a state of "data absence." Pain becomes an invisible, unspeakable private burden. Their bodies constitute one

The incidence of chronic musculoskeletal pain among rural women is as high as 75% (based on a preliminary survey by the local health clinic), but less than 30% of them have clearly described the location, intensity, and duration patterns of their pain to a doctor—because they lack tools for self-recording and channels to be heard.



C ■ Needle and Thread Memory: Embroidery Already Carries the Body

In the Miao communities of southeastern Guizhou, embroidery remains an integral part of many women's daily lives. As the needle rises and falls, it records not only patterns but also the passage of time, the rhythms of emotion, and the traces of the body:

- The density of stitches implicitly captures hours of handwork and physical fatigue.
- The depth of colour reflects sensory experiences of emotion and pain.
- The choice of floral motifs carries generational cultural meanings and bodily metaphors.

From these everyday practices we glimpse a possibility: embroidery itself can be a form of data encoding. If stitch, colour, and pattern are endowed with shared semantic rules, every single needle movement can become a deliberate record of bodily experience, rather than merely a decorative material practice.

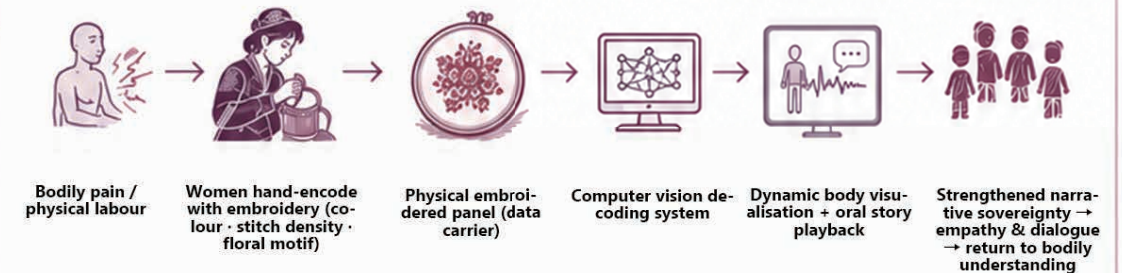
Traditional Miao embroidery:
decorative pattern → carrier of cultural memory

Data Embroidery:
encoding bodily pain, time, and labour → a legible personal health narrative

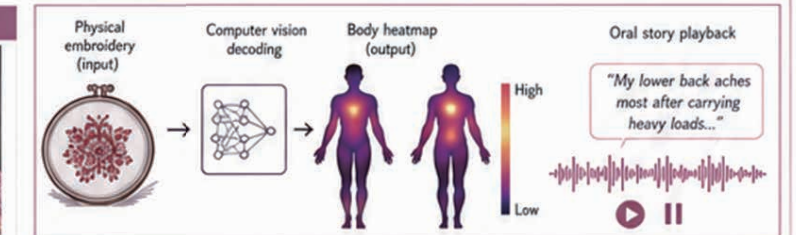
D ■ Our Response: Data Embroidery

Concept Definition

Data Embroidery is a design approach that reimagines traditional embroidery as a physical data-encoding medium. Rural women use needle and thread to hand-stitch their bodily labour and pain data into physical embroidered panels. Computer vision technology then deciphers the encoded data, transforming it into dynamic body visualisations and oral-story playback. It stitches together material craft and digital recognition, giving invisible pain a visible shape and an audible voice.



Encoding Logic Embedded in Traditional Embroidery



E ■ The Core Question We Address

How can rural women encode bodily labour and pain data through embroidery?

- Co-create an encoding language: work with rural women to jointly define a set of body-pain embroidery encoding rules based on colour, stitch density, and floral motifs.
- Create personal embroidered panels: each woman independently completes a personal data-embroidery panel based on one week of her own real bodily data.
- Develop an interactive system: build the computer-vision-driven decoding and narrative system XiuTrace, realising the experience of "touch a flower, hear a body's story."
- Establish a complete loop: form an autonomous narrative cycle of Body → Needle → Data → Interaction, enhancing women's sense of ownership over their own bodily data.

F ■ How can rural women encode bodily labour and pain data through embroidery?

- Location: A Miao village in the Qiandongnan Miao and Dong Autonomous Prefecture, Guizhou Province ("Embroidery Village"), with a living embroidery tradition and intensive physical labour.
 - Participants: 14 Miao women aged 32–68 (mean = 47.2), 12 of whom have long-term embroidery practice.
 - Collaborating institutions: Local ethnic cultural cooperative, Tsinghua University Academy of Arts & Design, Yunnan Minzu University School of Sociology.
 - Research duration: A five-week participatory design workshop series (scheduled outside farming peak seasons), supplemented by semi-structured interviews and system evaluation.
- Their stories begin with a single stitch.*



兰州大学威尔士学院
LANZHOU UNIVERSITY WALES COLLEGE

From Body Diary to Needle and Thread · 从身体日记到一针一线

An Embroidery Data Language Defined by Rural Women Themselves

Lanzhou University Wales College
Yuki



Scan for full paper

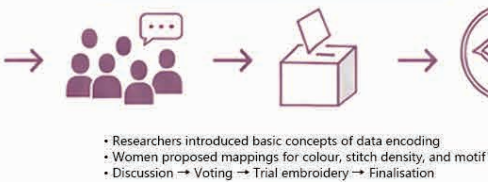
B ■ The Workshop Journey: Three Stages, Five Weeks Together

Week 1: Telling the Body



- Marked pain sites on body outline diagrams
- Told stories of the most memorable pain experiences from their labour
- Introduced a simplified "Body Pain Diary" and began one week of self-tracking

Weeks 2-3: Co-creating Encoding Rules



- Researchers introduced basic concepts of data encoding
- Women proposed mappings for colour, stitch density, and motif
- Discussion → Voting → Trial embroidery → Finalisation

Weeks 4-5: Crafting Personal Embroidered Panels



- Based on one week of each woman's own real bodily data
- Independently completed a 25 cm × 25 cm personal data-embroidery panel
- Recorded oral stories narrating the bodily memories behind the panels

Note: All workshops were held during gaps in farming work, each session lasting approximately 3 hours, at the village embroidery house.

C ■ Pain Has a Location, and a Story

The workshop did not begin with designing rules, but with "telling the body." Each woman received a blank human body outline diagram and was invited to circle the parts of her body where she felt pain most often, and to describe that feeling in her own words.



High-Frequency Pain Map

- Lower back**
"Like a stone tied on"
- Shoulder/back**
"Where the basket strap bites in"
- Wrists**
"Swelling from pinching tea buds tens of thousands of times"
- Knees**
"Can't stand up after squatting too long"

(Data from women's self-reports in the workshop)

Pain Diary: Building an Archive for the Body

The women then received a simplified "Body Pain Diary" form and were asked to record the following every day for one week, in their own way:

- Where did it hurt today?
- How strong was the pain? (Five levels from "a dull ache" to "hurts so much I can't sleep")
- How long did it last?

This diary later served as the raw data for encoding each embroidered panel.

D ■ Translating the Feeling of Pain into Needle and Thread

The researchers introduced the basic idea of data encoding: colours can represent numbers, stitch density can represent duration, and motifs can represent categories. Then the real agency was handed to the women. Drawing on their own bodily sensations, embroidery experience, and Miao cultural associations, they defined a shared set of encoding rules through discussion, voting, and trial stitching.

Data Dimension	Encoding Element	Encoding Scheme	Cultural Basis & Women's Explanation
Pain intensity (Levels 1-5)	Thread colour	1 · Light blue — gentle breeze	Women used natural colours as metaphors for bodily sensations: "The reds are like fire burning, like blood pulsing."
		2 · Pale green — sore & swollen	
		3 · Yellow — dull ache	
		4 · Orange-red — tugging pain	
		5 · Deep red/black-red — piercing pain	
Pain duration (hours/day)	Stitch density	Sparse running stitch — < 2 hours	"The denser the stitching, the longer the pain" — a natural metaphor between stitch density and accumulated time.
		Medium satin stitch — 2-5 hours	
		Dense seed stitch / knot stitch — > 5 hours	
Pain location	Floral motif	Head/neck → Peony	Morphological resemblance between flowers and body parts, plus the specific cultural semantics of flowers in Miao embroidery.
		Shoulders/back → Maple leaf (Miao tree-of-life image)	
		Lower back → Lotus	
		Wrists/hands → Plum blossom	
		Knees/legs → Chrysanthemum	
Labour type (auxiliary)	Background texture	Water ripple pattern — rice transplanting	Added by a few participants to incorporate work memory; optional, used to enrich the narrative.
		Grid pattern — weaving	
		Dot-line pattern — tea picking	
		Sawtooth pattern — firewood chopping	

Note: The "sparse/medium/dense" scale for stitch density was defined by each woman relative to her own everyday embroidery habits, preserving personal style. This design choice was later accommodated in the computer vision system through personalised calibration.

Five Floral Motifs & Body Parts



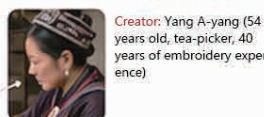
E ■ Fourteen Pieces of Cloth, Fourteen Body Narratives

After the encoding rules were established, each woman received a uniform 25 cm × 25 cm cotton-linen fabric, a silk thread set, and her own one-week body diary. They independently created their personal data-embroidery panels based on their real data, with a creation period of ten days. Researchers provided only technical assistance, never interfering with content or composition. All 14 women completed their panels.

Full Interpretation of One Panel



The Week of Picking Qingming Tea



- Deep red lotus (centre of the panel)**
— Lower back pain, intensity level 5, > 5 hours, densely filled seed stitch. "My back hurts the most, so the lotus is the biggest and the darkest."
- Pale green plum blossom (lower right)**
— Wrist soreness, intensity level 2, 2-5 hours, satin stitch. "Pinching tea buds tens of thousands of times makes my wrists sore, but not the worst pain."
- Light blue peony (upper area)**
— Mild headache, intensity level 1, < 2 hours, sparse running stitch. "It rained those days, my head ached a little, but I could still work."
- Dot-line background texture**
— Tea-picking labour.

Excerpt of Oral Story

"This plum blossom on my wrist was stitched for pinching the tea buds... tens of thousands of pinches. The lotus on my lower back was stitched for all that bending. I filled it with the densest seed stitch, just as that pain filled the whole day. When I was embroidering, the needle going in made me feel the pain again. But I also felt, at last, that someone could see my pain."

- 14 Embroidered Panels
- 5 Floral Motifs
- 5 Colour Gradient Levels
- 3 Stitch Density Grades

All hand-stitched by the women themselves, no two alike

More Embroidered Panels



F ■ In Their Own Words...

- "Before, when I told men my back hurt, they thought I was being fussy. Now that I've stitched it out and they see it, they finally go 'oh.'"
- "So our old stitches can record all this—now even the younger ones want to learn."
- "As I stitched, I counted the stitches, just like I counted how many times I bent over that day."



Discussing colour samples and encoding rules together



A woman shares her just-finished panel.



Researcher provides technical support without interfering.

References

- [1] Folten, L. et al. Pain and women's work in agrarian economies. *Sex So Med*. 2019;22(6):1-10.
- [2] Ingrid T. Making. Anthropology, archaeology, art and architecture. Routledge; 2013.
- [3] UNESCO. Intangible Cultural Heritage in traditional textile craftsmanship. 2017.
- [4] Lingmei D. The craft of embroidery. *Innovation*. 2021;2(6):514-41.
- [5] Zhao Y. et al. Data-embroidery: a new medium for personal health storytelling. *Digit Health*. 2023;9:1-12.



兰州大学威尔士学院
LANZHOU UNIVERSITY WALES COLLEGE

XiuTrace: Touch a Flower, Hear a Body's Story · 绣迹：触摸一朵花，听见身体的故事

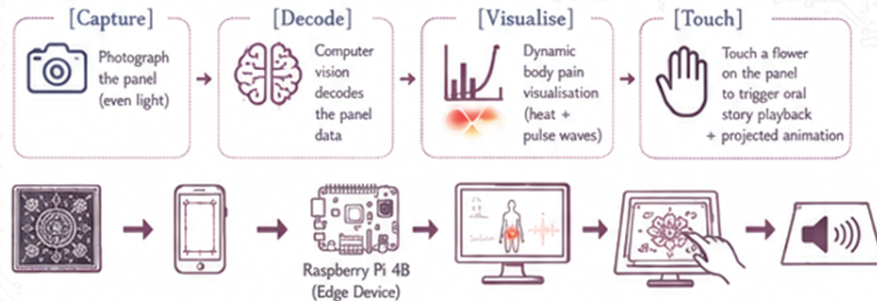
A Physical Data Interaction System Weaving Together Computer Vision and Miao Embroidery Tradition

When women stitch their bodily pain into fabric, how can these "data-embroidered panels" be read by machines?
And how can that reading be guided back toward the body to evoke empathy?
We developed the "XiuTrace" system, which allows invisible pain to regain shape and voice through touch and vision.



Scan for full paper

B ■ Invisible Bodies: Rural Women's "Data Absence"



Core Loop Description

- Capture** — The user photographs the front of the embroidered panel with a smartphone inside an even-light box.
- Decode** — The image is sent to a local Raspberry Pi 4B running a custom vision pipeline that performs colour segmentation, floral motif recognition, and stitch density estimation, outputting structured body data.
- Visualise** — A web-based interface generates an abstract female body outline; pain sites "heat up" and change colour according to intensity, and produce spreading pulse waves according to duration.
- Touch** — The panel is placed on a transparent touch-sensing surface. When a viewer touches a flower on the panel, the system projects an enlarged dynamic visualisation and plays the oral story of the person who stitched that pain site.

C ■ How Does Computer Vision "Read" Embroidery?

The embroidered encodings are not cold, standardised data but handcrafted traces bearing personal styles. To enable the machine to decode accurately while respecting cultural differences, we designed four connected recognition modules and introduced a personalised calibration mechanism.

- 1 Colour Segmentation & Pain Intensity**
K-means clustering (K=6) in HSV colour space segments the embroidered surface into different colour regions. A predefined colour lookup table maps each region to a pain intensity level of 1-5. Personalisation design: On first use, each creator stitches a small "colour calibration card," enabling the system to build a user-specific colour-intensity mapping model and eliminate individual dyeing variations.
- 2 Floral Motif Recognition & Body Site Mapping**
We collected panels from 14 participants, manually annotated 210 floral instances, applied data augmentation, and fine-tuned a MobileNetV2 convolutional neural network. The network recognises five motif categories — peony, maple leaf, lotus, plum blossom, chrysanthemum — mapping them to head/neck, shoulders/back, lower back, wrists/hands, and knees/legs respectively.
- 3 Stitch Density Estimation & Duration Classification**
Pattern regions are converted to grayscale; Gaussian filtering and Local Binary Pattern (LBP) texture feature extraction are applied, using the equivalent texture contrast as a quantitative index of stitch density. The result is compared against each creator's personal calibration scale (sparse/medium/dense), and classified into three intervals corresponding to <2 h, 2-5 h, and >5 h.

- Template matching detects background textures — water ripples, grids, dot-lines, sawtooth patterns — to add labour-type labels and enrich the narrative context.
- Local processing (Raspberry Pi 4B) to protect privacy
- Personalised colour + density calibration, respecting cultural differences
- Lightweight CNN (MobileNetV2) runnable in low-resource settings



D ■ Interaction Design: Guiding Data Back to the Body

The decoded numbers are not the endpoint. Our interaction design principle is — feedback must lead back to the body, not stop at abstract charts. To achieve this, we designed a dual-channel interaction: dynamic body visualisation and touch-triggered oral narrative.

1 Dynamic Body Visualisation

A web interface displays a female body outline drawn in a Miao paper-cut style. Each body part "heats up" and shifts colour according to the decoded pain intensity — from light blue (no/mild pain) to deep red (severe pain). Around the pain site, dynamic pulse waves are generated: the longer the duration, the slower the pulse frequency, but the larger the diffusion radius — a metaphor for chronic, lingering pain. Clicking on a body part reveals a numerical summary and the distribution of pain days.



2 Touch-Based Narrative Interaction

The embroidered panel is placed on a transparent conductive grid touchpad (modified from an infrared touch frame). When a viewer touches a floral motif on the panel, the system maps the coordinates to the corresponding body site. A projector casts an enlarged dynamic visualisation of that site beside the panel. The system simultaneously plays back the oral story audio recorded by the panel's creator for that site.

"We did not want pain to become a set of cold numbers. We wanted, when a viewer touches the panel, for them to feel the density of the stitches, the warmth of the colours, and to hear the person who stitched it all tell the story of her body."

An elderly female viewer gently touches the deep-red lotus on a panel; beside it, a projection screen immediately lights up with an animation of a body with a red-glowing waist and slowly expanding pulse waves, while the creator's voice begins: "This lotus is my lower back, filled with the densest seed stitch, just as that pain filled the whole day...restor for that site."

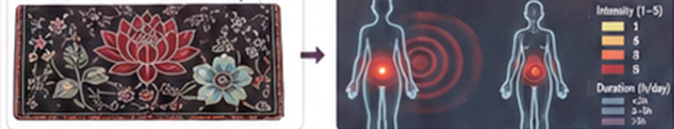
E ■ System Evaluation: 14 Creators + 12 Village Members

We conducted a two-stage evaluation:
Internal evaluation: 14 panel creators interacted with their own and others' panels in turn, then completed questionnaires and interviews.
External evaluation: 12 women from the same village who had not participated in designing the encoding rules were told only that "secrets of the body are hidden in the panels." They interpreted the panels by touching them and viewing the visualisations. We recorded their interpretation accuracy and emotional responses.

Metric	Result	Interpretation
Overall decoding accuracy	87.3% (SD=6.1%)	External villagers could interpret with high accuracy without any training.
Body-site recognition accuracy	94.2%	Extremely strong consensus on the mapping between flowers and body sites.
Intensity recognition accuracy	85.6%	Good recognition of the colour gradient.
Duration recognition accuracy	80.2%	Individual variation in stitch density caused some boundary blurring.
System Usability Scale (SUS)	82.4 (SD=9.7)	"Excellent" rating.
Bodily Empathy Resonance	5.87/7 (p < .001)	Significantly above the mid-point; highest items: "I feel this person's bodily experience is worth listening to" (6.41) and "This panel reminds me of my own body's pain" (6.23).

- "When I saw that deep-red lotus, it was like seeing myself... so everyone's backs are ruined." (55-year-old participant, in tears upon seeing another woman's panel)
- "Before, when I told men my back hurt, they thought I was being fussy. Now that I've stitched it out and they see it, they finally go 'oh.' "traction
- "Touching the flower to hear the story—it's like the flower is speaking." (External evaluation participant) Extraction

From Panel to Visualisation: One Example



G ■ System Evaluation: 14 Creators + 12 Village Members

- Proposed the concept of Data Embroidery, establishing traditional handicraft as a computationally legible physical data-encoding medium, with encoding performed by the data subject herself.
- Proposed the concept of Data Embroidery, establishing traditional handicraft as a computationally legible physical data-encoding medium, with encoding performed by the data subject herself.
- Proposed the concept of Data Embroidery, establishing traditional handicraft as a computationally legible physical data-encoding medium, with encoding performed by the data subject herself.

F ■ The Changes Brought by the Embroidered Panels: Three

Through thematic analysis of the interviews, we distilled three core findings that reveal the social and emotional impact of the Data Embroidery system beyond its functionality.

Theme 1: "The pain has been seen"

— From Invisible Pain to Narrative Sovereignty
The women emphasised that the fundamental value of the panels and system lies not in precise measurement but in making invisible pain visible, touchable, and discussable. The materialised data became a discursive tool for women to seek family understanding and self-care.

Theme 2: "It hurt while stitching; I understand while looking" — The Empathic Power of Embodied Data

The bodily memories of the creators were reactivated during the stitching process, and viewers, through touch and visual perception, experienced strong embodied empathy. The density of the stitches was directly felt as the weight of time; the warmth of the colours was perceived as the burning sensation of pain. Data was transmitted directly from body to body, bypassing the intermediary of abstract numbers.

Theme 3: "Old stitches, new purpose" — Revaluing Traditional Craft

Elderly embroiderers felt their craft had been endowed with new public significance. "I never imagined our old stitches could record this kind of data—now even the younger ones are interested in learning." The system became a bridge for transmitting bodily knowledge and embroidery skills across generations.

H ■ Conclusion and Outlook

From body to needle and thread, from needle and thread to data, and from data back to the body —
Data Embroidery stitches together not only information, but also the pain that has been neglected, the narratives that have been silenced, and the bodies that have been forgotten.

Future Directions

- Multi-village comparative studies to explore the cultural diversity of encoding rules
- Integration with community health services to explore the auxiliary role of Data Embroidery in geriatric chronic pain management
- Introducing distributed ledger technology so that embroidered panels can serve as trustworthy bodily data credentials
- Open-sourcing the visual decoding toolkit to enable more communities to create their own "embroidery data languages"

NO.6

Research Question 1: In cross-cultural co-design, how do craft practitioners' "embodied metaphors" negotiate and reconfigure the standardized encoding logic of technological systems?

WHY IS THIS A GOOD QUESTION?

This question does not ask vaguely whether "metaphors are useful" but precisely targets a tension-filled dynamic process—"negotiation" and "reconfiguration." It presupposes that craft knowledge is not a passive "data source" but an epistemological system with its own logic. The question is superior to a general one because:

Process-oriented: It asks "how," requiring the research to focus on the concrete mechanisms through which metaphors emerge, collide, fail, or achieve consensus, rather than merely reporting a binary result of "success/failure." This aligns perfectly with the two contrasting cases in the paper: "Dense stitching means long-lasting pain" and the generational divergence over color.

Epistemological critique: It presupposes that the "standardized technological logic" is itself part of the problem, not the sole correct answer. The aim of the research is not to make embroiderers adapt to a standard, but to observe how their knowledge "reconfigures" that standard. This provides a legitimate theoretical landing point for Pan Jie's fundamental challenge: "If I stitch it your way, will it still be my back pain?"

Transferability potential: The answer can transcend the embroidery context and offer a "methodological tool" for any field involving the integration of local knowledge and technical systems (e.g., traditional medicine and public health, local ecology and climate science)—namely, how to identify and make good use of metaphors rooted in bodily practice to build more equitable collaborations.

Research Question 2: To what extent can incorporating "the refusal of standardization" and "individual differences" into a design scheme address epistemic injustice, and what new ethical dilemmas might this create?

WHY IS THIS A GOOD QUESTION?

The value of this question lies in its intensely charged, dilemmatic structure. It refuses to simply celebrate the research and instead forcibly elevates "ethical reflection" from a subsidiary part of the study to a core exploratory goal.

A complete circuit from "justice" to "cost": It first asks about the effectiveness of "addressing epistemic injustice," which will guide the research to examine whether the "personal calibration" approach truly gave the embroiderers narrative sovereignty. More importantly, it asks "what new predicaments are created," compelling the researcher to confront issues like the sacrifice of "cross-individual commensurability" in public health assessment, the "sustainability burden" brought by the increased system complexity intended to respect difference, and the potential for creating new forms of "technological exclusion."

Institutionalizing reflexivity: This question itself serves as a constant vigilance against researcher-centeredness. It requires the researcher, throughout the entire design process, to view every empowering technical decision as simultaneously an exercise of power that may cause harm, thereby solidifying self-reflection from a momentary insight into a persistent dimension of exploration.

Promoting "honest" knowledge production: Against the backdrop of participatory design often being idealized, this question encourages reporting "failures," "costs," and "unresolved tensions," just as the paper itself does. This helps the academic community move beyond "harmony narratives" and engage in more grounded and credible knowledge accumulation.

Research Question 3: During the translation and knowledge transduction processes of co-design, how are the researchers' own epistemological presuppositions exposed, challenged, and revised?

WHY IS THIS A GOOD QUESTION?

This is a courageous question that completely turns the lens around onto the researchers themselves. It pulls the researcher down from the position of an "objective observer" and makes them one of the primary subjects of the study.

Operationalizing "epistemic humility": This question no longer talks abstractly about "researcher positionality" but instead demands using empirical data (such as workshop transcripts, moments of being "politely corrected" by the embroiderers, the revision history of drafts that repeatedly slip back into over-theorization during writing) to demonstrate exactly "how" the researchers' presuppositions were peeled back and revised layer by layer. This makes reflexivity visible and analyzable.

Revealing micro-power dynamics in knowledge production: By focusing on the concrete acts of "translation" and "transduction," this question can uncover a subtle power process: how researchers, in the name of "understanding," may over-theorize the embroiderers' simple bodily experiences and package them as academic outputs. Pan Jie's remark, "You teachers really do have such fancy ideas," serves as the core cautionary note for this question.

Providing a learnable methodological pathway: The answer to this question is not a set of moral platitudes, but a concrete set of methodological practices—such as how to conduct reflexive discussions, how to retain and analyze field notes from moments of being "put in one's place," and how to handle the loss of meaning in multilingual writing. For other scholars wishing to engage in responsible participatory research, this is an invaluable operational framework that can be borrowed and critiqued.

RESEARCH QUESTIONS

NO.7

Practice-Based Research Proposal Outline

Project Title: The Stitches of Metaphor: A Practice-Based Study on Embroidered Body Data and Cross-Cultural Co-design

Researcher: Yuki

Field: Participatory Design / Data Physicalization / Craft Knowledge

实践型研究提案大纲
项目名称： 隐喻的针脚： 一项关于身体数据刺绣与跨文化共同设计的实践型研究
研究者： 雨琦
领域： 参与式设计 / 数据物理化 / 手工艺知识

(A) RESEARCH QUESTIONS

- 1.The negotiation mechanism of embodied metaphors: In the process of encoding bodily pain data through embroidery, how do the embodied metaphors spontaneously generated by Miao embroiderers collide with the standardized technological logic presupposed by the designers, and how do they ultimately reshape the encoding rules?
- 2.The ethical cost of framing individual difference as a cultural resource: By incorporating the embroiderers' "refusal of standardization" and their individual stitch variations into a design solution (the personal calibration system), to what extent can epistemic injustice be addressed? And what new ethical dilemmas does this choice create (e.g., cross-individual incommensurability, burdens on technological sustainability, potential exclusion)?
- 3.The epistemological revision of the researcher: During the processes of co-design and cross-linguistic knowledge transduction, how are the researchers' own theoretical presuppositions exposed, challenged, and revised by the participants' concrete bodily knowledge? How can this revision be documented and transformed into an actionable methodological vigilance?

(B) BACKGROUND/CONTEXT

This study takes place in a Miao village in Qiandongnan, Guizhou, which maintains a complete tradition of Miao embroidery. Local middle-aged and elderly women commonly engage in heavy manual labor and suffer from chronic pain, yet their bodily experiences have long lacked a legitimate form of expression within mainstream medical scales and standardized data systems—this is precisely what Fricker terms "hermeneutical injustice." Embroidery, as the material practice most familiar in their daily lives, embodies a rich craft epistemology: the tightness of stitches and the warmth or coolness of colors are not merely technical parameters, but materialized marks of personal emotion, bodily state, and style. The starting point of this research is a moment of "standardization collision" during a co-design workshop: when the researchers attempted to establish a unified mapping standard between stitch density and pain duration, one embroiderer rejected this plan with the words "every family's stitch tension is different," thereby opening an alternative path that uses embodied metaphors as tools for negotiation and personal calibration as a technical outlet. This study is a deepening and re-practice of that process and its aftershocks. This research continues the intersecting lineage of participatory design, data physicalization, and craft studies, but places greater emphasis on integrating the researcher's reflexivity into the practice itself, making the research process a sustained and visible epistemological self-correction.

(C) RESEARCH METHODS

This study adopts a practice-based research framework, with co-design as the core method, supplemented by ethnographic observation, auto-ethnographic documentation, and reflexive analysis. The specific steps are as follows:
1.Relationship Building and Field Immersion
Enter the community through the local cultural cooperative, participate in daily embroidery activities, and build trust relationships that transcend the "project." This phase simultaneously involves participant observation, documenting the embroiderers' everyday expressions regarding pain and embroidery.

2.Co-design Workshops
Round 1: Morphological metaphor mapping between body parts and floral motifs (waist → lotus, etc.).
Round 2: Negotiating the encoding of stitch density and pain duration, allowing metaphors to emerge spontaneously, and documenting consensus and conflict.
Round 3: Synesthetic metaphor negotiation between thread color and pain intensity, with a special focus on generational divergences.
Round 4: Creating personal calibration cards—each embroiderer embroiders sample patches using her own definition of "sparse," "medium," and "dense," serving as a personal density scale.

3.Creation of Personal Embroidered Pieces and Oral Narrative Recording
Based on a one-week pain diary, each embroiderer creates her own data-embroidered piece using the negotiated encoding rules. The creation process is video-recorded, and their explanations of the work and pain narratives are recorded afterward. This process itself constitutes embodied reenactment.

4.Co-development of an Interactive Prototype
In collaboration with technicians, develop an interactive system where touching the floral motifs on the embroidered pieces triggers corresponding bodily pain animations and the maker's oral stories. The system uses a personal calibration model, learning the stitch and color characteristics of each embroiderer, and rejecting unified thresholds.

5.Auto-ethnographic Practice
Write reflexive field notes after each workshop, focusing on: Where were my presuppositions challenged? Am I over-theorizing the participants' spontaneous expressions?
Keep all drafts and revision traces, tracking the moments where I "repeatedly slip into theorization and am pulled back" during analysis and writing.
Conduct regular reflexive discussions with the second and third authors, documenting the loss of meaning and choices made during translation.
Encode the moments of being "politely corrected by the embroiderers" as key data for analysis.

6.Analysis and Integration
Conduct thematic analysis on workshop videos, interview transcripts, images of embroidered pieces, and reflexive notes. The analytical framework includes: conditions for the success or failure of metaphor negotiation, ethical consequences of personal calibration, and the trajectory of change in the researchers' epistemological presuppositions.

PBR PROPOSAL OUTLINE

(D) ANTICIPATED OUTCOMES

Practical Works/Artifacts:
14+ personal data-embroidered pieces: Each participating embroiderer creates 1-2 pieces, materializing her bodily pain narrative in silk thread and cloth.
An archive of personal calibration cards: Containing the density sample patches and color calibration cards of all embroiderers, serving as a "personal bodily encoding dictionary."
An interactive installation prototype: Viewers touch the floral motifs on the embroidered pieces, triggering corresponding body pain animations and the maker's oral stories. The system embeds the personal calibration model, learning from individual craft variations.
A process documentary film: Edited footage of core moments from the workshops, creation, and oral narratives, presenting the conflicts, negotiations, and repairs within the co-design process.

Knowledge Outputs:
A reflexive methodological paper: Systematically reporting on the effectiveness and boundaries of metaphors as negotiation tools, the ethical costs of the personal calibration approach, and the process and mechanisms of the researchers' own epistemological revision.
A transferable framework of co-design principles: Distilling principles such as "metaphor builds consensus, personal calibration catches the differences," and the epistemological stance of "refusal as data," for reference by cross-cultural co-design projects in fields like health, ecology, and cultural heritage.
A researcher's self-vigilance checklist: Based on the recurring tendency toward over-theorization in this study, summarizing a set of operational points for reflexive practice, such as "maintain a suspension between academic concepts and bodily language" and "prioritize preserving the participants' own explanatory frameworks."

KEY QUESTIONS FOR SELF-RESPONSE:

What is your creative domain/field?
My field is the everyday embroidery space in a Miao village in Guizhou and the site of the co-design workshops. It is the physical village and the embroidery frame, as well as the epistemological contact zone between two knowledge systems (craft knowledge and technological rationality).

What are your research questions?
See (a) above. The three questions focus respectively on the mechanism of metaphor negotiation, the ethical costs, and the researcher's self-correction.

What practice-based research methods will you use?
Co-design as the core, integrated with participant observation, auto-ethnography, reflexive writing, and thematic analysis. My practice does not lie in unilateral"making," but in co-creating encoding rules, embroidered pieces, and the interactive system with the embroiderers, and in making my own reflective process an object of study.

What works/artifacts will you create? Describe the creative process.
The core works are data-embroidered pieces and an interactive installation. The creation process of the embroidered pieces is independently completed by the embroiderers based on their pain diaries and using the personally calibrated encoding rules; it is the materialization of their bodily knowledge. My creative intervention lies mainly on the side of the interactive installation—embedding the personal calibration logic into the system design, so that the machine becomes a learner of individual differences.

How will you integrate auto-ethnographic practice? How will you document the process?
Through reflexive field notes, recordings of team discussions, and version tracking of writing drafts. Special attention will be paid to moments of "being corrected," and the repeated process of slipping into academic discourse during theorization and pulling back. These materials themselves constitute data.

What are the relevant theories? Who are the key theorists involved?
Fricke's"epistemic injustice"and"hermeneutical injustice,"Lakoff and Johnson's"metaphor theory,"Sennett's and Ingold's"craft epistemology,"Light and Akama's work on"discomfort and conflict in participatory design."

What analytical method will you use for your own work and comparative works?
Thematic analysis will be used to encode process data, constructing categories for the conditions of success/failure in metaphor negotiation, types of ethical costs, and stages of reflexive transformation. For comparison, my initial standardized scheme will be compared with the final agreement corrected by the embroiderers, and the different understandings of the same encoding dimension among different embroiderers will be compared, to reveal how individual differences transform from "errors" into "cultural resources."

NO.8

When Embroiderers Refuse Standardisation: Metaphor Negotiation in Co-designing Embroidered Body Data with Rural Women

当绣娘拒绝标准化：数据刺绣共同设计中的隐喻协商

ABSTRACT

Body data practices in marginalised communities often fail due to externally imposed standardisation frameworks — this is not only a technical problem but a form of epistemic injustice. Through a co-design study on data embroidery with rural women in Yuzhong, Lanzhou, this paper examines how, when craft holders are invited to become co-creators of data encoding rules, their embodied knowledge collides with the standardising assumptions of technical rationality and ultimately recalibrates the latter’s epistemological premises. The study finds that the key mechanism for reaching encoding consensus is embodied metaphor rooted in bodily experience — yet metaphor does not always succeed; it breaks down when different generations hold divergent sensory frameworks for the colour of pain. The embroiderers’ insistence on personal stitch style constitutes a form of “refusal” of data standardisation, compelling the designers to reframe individual difference from technical error to cultural space. The researchers themselves also underwent epistemic recalibration in this process — a moment of polite correction by an embroiderer made us realise how readily we over-theorise participants’ improvisational expressions, and this reflection is not a one-off event but a process of repeatedly slipping into and being pulled back from abstraction throughout research and writing. The core contributions of this paper are threefold: it reveals the efficacy and boundaries of metaphor as a negotiation tool in cross-cultural co-design; it demonstrates that craft epistemology can, at the level of system design, address what Fricker terms hermeneutical injustice; and it candidly discusses the ethical costs of designing individual difference as a cultural resource along the two dimensions of technological sustainability and cross-individual incommensurability. Although the specific context of this study is data encoding and embroidery craft, these findings offer transferable methodological insights for any co-design project involving negotiation across knowledge systems — especially in the contexts of health, body narratives, and marginalised communities.

KEYWORDS:

co-design; epistemic justice; metaphor negotiation; craft knowledge; body data; cross-cultural collaboration

1. Introduction: The Standardisation Collision

On the third day of the co-design workshop, an unforeseen impasse emerged.

We were attempting to co-develop, with fourteen embroiderers in Lanzhou, a set of encoding rules for “embroidering” bodily pain data. The first two days had progressed unexpectedly smoothly: the mapping of pain sites to floral motifs (waist → lotus, head → peony) was rapidly agreed upon due to morphological similarity and Miao embroidery traditions. However, when we began discussing how to represent the duration of pain through stitch density, an older embroiderer set down her needle and thread and spoke a sentence in the Miao language that suddenly chilled the atmosphere: “Every family’s stitch tightness is different. If you make me embroider according to your pattern, will what comes out still be my own backache?”

Her question struck at the core tension of the entire project: as designers, we arrived with the expectation of establishing a set of encoding standards that would be “universally applicable and accurately machine-readable”; as craft holders, they insisted that the personal nature of stitching is an inseparable part of bodily narrative. To standardise “dense” and “sparse” into uniform numerical values was, in their view, to strip the experience of pain from their own bodies and hands.

We call this moment the “standardisation collision.” It is not a mere “miscommunication” that can be easily smoothed over, but an epistemological rupture — one side pursuing machine readability, the other defending the authenticity of bodily expression [6]. How this collision is handled determines the direction of the entire collaboration: whether to repeat the “manufacturing consent” trap repeatedly warned against in participatory design literature, or to truly realise the knowledge equality promised by co-design [4].

This paper is an in-depth reflection on this collision, its repair process — and the new problems exposed after the repair. We report on a week-long data embroidery co-design study conducted in Yuzhong, Lanzhou. Unlike research that takes system output as its endpoint, this paper focuses on the epistemic politics of the co-design process itself. We ask: When encoding rules must be generated through cross-cultural, cross-knowledge-system negotiation, how does craft practitioners’ bodily knowledge challenge and reshape the standardising logic of technical rationality? What role does metaphor play in bridging epistemological ruptures — and what happens when metaphor fails? What ethical costs do we pay in the process of designing individual difference as a cultural resource? And — equally importantly — what epistemic recalibration do the researchers’ own presuppositions undergo during this process?

2. Theoretical Framework: Epistemic Injustice and Craft Knowledge in Co-design

2.1 Hermeneutical Injustice: When Bodily Pain Lacks Legitimate Forms of Expression
Fricker (2007) distinguishes two types of epistemic injustice. Testimonial injustice occurs when a person’s words are disbelieved due to prejudice — for example, when rural women’s chronic pain is dismissed by family or doctors as “affectation” or “which woman doesn’t feel like this.” Hermeneutical injustice is more insidious: it occurs when a group lacks the conceptual tools to interpret their own experiences. When a hospital pain scale requires patients to rate pain on a 1–10 scale, and an embroiderer who has never encountered abstract scales can only describe it as “a dull ache,” “a gripping pain,” or “hurts so much I can’t sleep,” what she faces is not merely a communication barrier but a structural knowledge exclusion — her experience is judged as “imprecise” and therefore unrecordable and untraceable within formal medical and statistical systems.

One of the core promises of co-design is precisely to create conditions for marginalised communities to “co-produce knowledge.” However, when designers enter communities with pre-packaged standardisation tools, we may inadvertently reproduce hermeneutical injustice — replacing local sensory vocabularies with external scales, and bodily truth with technical precision. Therefore, achieving epistemic justice does not lie in “teaching communities to use better tools,” but in creating a space where local knowledge can calibrate the epistemological premises of external tools [3]. As Escobar (2018) argues in his discussion of designs for the pluriverse, the task of design is not to translate multiple realities into a single framework, but to support their existence and recognition on their own terms.

2.2 Craft as Epistemology

Sennett (2008), in his classic work on the craftsman, points out that craft knowledge is quintessentially “tacit knowledge” — it cannot be fully expressed in propositional language but is internalised through years of bodily practice and repeated interaction with materials. Ingold (2013) further proposes that making itself is a way of knowing: the craftsperson does not first design and then execute, but generates form and meaning through continuous dialogue with materials. Miao embroidery tradition is precisely such an epistemological system: stitch tightness is not merely a technical parameter, but a materialised imprint of the embroiderer’s mood, bodily state, and personal style. When our collaborator insisted that “every family’s stitch tightness is different,” she was asserting the legitimacy of this craft epistemology — data should not be abstracted away from the making process, because the making process itself is part of the bodily narrative.

This stance challenges the usual practice in the field of data physicalisation. Existing research typically has designers unilaterally define data-to-material mapping rules, with craft practitioners serving only as executors [1][2]. Our study attempts to invert this logic, but the process of inversion exposes the deep friction between two epistemologies — technical rationality and craft knowledge.

PAPER OUTPUT

2.3 Metaphor as a Cross-Cultural Negotiation Tool and Its Boundaries*

Lakoff and Johnson (1980) propose that metaphor is not merely rhetoric, but a fundamental cognitive mechanism through which humans understand abstract concepts. People understand abstract target domains (such as pain, time, emotion) through concrete, embodied source domains (such as temperature, space, touch). In cross-cultural co-design, this theory holds significant methodological potential: when abstract technical terms lack cultural equivalents, metaphors grounded in shared bodily experience may serve as bridges for connection (Akama and Prendiville 2013).

Yet metaphor is not all-powerful. Lakoff and Johnson also note that metaphorical mapping is deeply influenced by cultural experience and individual differences. Different cultures may use different source domains to understand the same target domain, and different generations or individuals within the same culture may also hold divergent views. In co-design literature, reports on metaphor failure are far fewer than success stories — which may itself be a publication bias. As Lee (2021) criticises in the journal CoDesign regarding “conflict avoidance” in participatory design, we often tend to report harmonious consensus processes while downplaying friction. This paper seeks to address this gap: we report both moments where metaphor successfully facilitated consensus and candidly present a situation where it failed amid generational divergence [5].

3. Research Context and Methods

3.1 Field Site and Partnership

The research took place in a village in Yuzhong, Lanzhou, which retains intact embroidery traditions and farming practices. The partnership did not start from scratch. The authors had already established preliminary trust through a cooperative where they assisted with product design. Such long-term relationships are crucial in participatory design — as Smith, Iversen, and Hjerimitslev (2017) emphasise, trust is not established once and for all with an “informed consent form” at the project’s outset, but accumulated through sustained co-presence and co-working. The study was approved by the university ethics committee, and all participants signed informed consent forms, retaining the right to withdraw at any time without affecting their standing in the cooperative.

3.2 Participants

Fourteen women participated voluntarily, aged 32 to 68 (median 47). Twelve had over 20 years of embroidery experience; two were young married women learning embroidery. All engaged daily in heavy physical labour such as tea-picking and rice transplanting, and all self-reported varying degrees of chronic lower back or joint pain. To protect privacy, all names in the text are pseudonyms, and separate permission was obtained for the use of embroidery images.

3.3 Researcher Positionality and Reflexive Practice

We positioned ourselves as “designers-as-learners.” The workshop facilitation and translation were primarily conducted in the Miao language by the second author Li Minghui (a Miao sociologist) and Wang Qiaohua; the first author was responsible for documentation and assistance. After each workshop session, the team held reflexive discussions, recording where our presuppositions were challenged by the participants’ words and actions — these records constitute a key data source for our analysis.

Here we must candidly acknowledge a limitation that ran throughout the process: language. The first author Zhang Xiaohui does not understand the Miao language, and all communication with older embroiderers (some of whom do not speak Mandarin Chinese) relied on translation by Li Minghui and Wang Qiaohua. This means that the embroiderers’ words as presented in this paper have already undergone a two-layer translation process: Miao → Mandarin → academic writing. Choices, simplifications, and interpretations in translation are unavoidable. We mitigated the risk of distortion by back-translating key quotations for participant confirmation, but this structural mediation is always present. In cross-linguistic co-design, such mediation is not an “impurity” to be optimised away, but an epistemological condition that researchers must honestly annotate (see Kim 2022 for a discussion of cross-language participatory research).

4. The Co-design Process: Negotiation, Collision, and the Successes and Failures of Metaphor

4.1 Smooth Consensus: Morphological Metaphor and Cultural Symbols

The encoding negotiation began remarkably smoothly. When asked “what motif would you use to represent backache,” the embroiderers almost unanimously chose the lotus, reasoning that “the waist is soft, and the lotus petals are curved.” The head was associated with the peony (“the head is important, the peony is the king of flowers”), and the hand with the plum blossom (“small little flower, like a fingertip”). The maple leaf — which symbolises clan origins in Miao embroidery — was used to represent the shoulder, because “the shoulder carries the whole family.”

From the perspective of Lakoff and Johnson’s classification framework, what was at play in this phase was a morphological metaphor: mappings were established using the visual morphological similarity between flowers and body parts. At the same time, the existing symbolic system in Miao culture (the peony’s nobility, the maple leaf’s clan significance) provided cultural anchorage for these mappings. The whole process saw almost no dispute, because morphological similarity and established cultural symbols provided a strong shared source domain here. This confirms the observation of Sanders and Stappers (2008): when the “fuzzy front end” of co-design has a sufficient basis of cultural consensus, negotiation can be highly efficient.

4.2 Collision: Stitch Density and the Rupture of Standardisation Assumptions

The conflict occurred when discussing how to represent the duration of pain. We proposed dividing daily cumulative pain duration into three tiers (<2 hours, 2–5 hours, >5 hours) corresponding to three stitch densities — sparse running stitch, medium satin stitch, dense seed stitch — and prepared reference samples demonstrating what we understood as “standard sparse,” “standard medium,” and “standard dense.”

At this point, the 54-year-old embroiderer Sister Pan voiced the challenge quoted at the opening of this paper. Subsequently, a younger embroiderer added: “My hand is tight, hers [pointing to an older embroiderer] is loose. My ‘sparse’ might be about the same as her ‘medium.’ If we have to follow one standard, whose counts?” The workshop fell into a brief silence.

This moment became the turning point of the entire study because it exposed a fundamental divergence between two epistemologies in their definitions of “precision.” Our “standard” attempted to reduce stitch density to an objective variable comparable across individuals. The embroiderers’ “personal standard,” however, was an embodied baseline rooted in long-term bodily habits — within their epistemological framework, differences in hand style are not noise to be eliminated, but the inseparable signature of individual bodily narrative. Sister Pan’s objection was, in essence, an intuitive resistance to what Fricker terms hermeneutical injustice: do not measure my body with your ruler.

From the perspective of co-design methodology, this moment also validates the warning of Light and Akama (2012): when designers enter a community with pre-formulated plans, even if we claim to be open, participants’ space for choice is already effectively constrained within a framework of accept or fine-tune.” Sister Pan’s refusal was not a fine-tuning of the proposal, but a fundamental challenge to the premise of the whole framework — the premise that stitch density needs to be standardised.

4.3 Repair: Emergence of Ontological Metaphor and Institutionalisation of Personal Calibration

Facing the impasse, we did not forcefully push for standardisation. Li Minghui asked in Miao: “Then how can you let others know that it hurt here for a little while, or that it hurt all day long?”

Sister Pan thought for a moment, pointed to a piece of seed stitch she was working on her embroidery frame, and said: “Look at these seeds, piled densely together — isn’t it just like pain that lingers a very long time and refuses to leave?” Another embroiderer pointed to a nearby area of sparse running stitch and said: “These stitches walk fast, like pain that only lasts a moment.”

“Embroidered dense means painful for long” — this offhand metaphor instantly won everyone’s nodding approval. In the theoretical framework of Lakoff and Johnson, this is a typical ontological metaphor: the abstract experience of time (pain duration) is reified into a tangible material accumulation (the compactness of stitches). “Refuses to leave” further strengthens the metaphor — pain is endowed with agency, becoming an unwelcome guest in the body, and the dense seed stitch externalises this experience of invasion in a visual and tactile way.

This metaphor grew purely from the bodily experience of embroidery. It echoes Sanders and Stappers’s discussion of the critical role of “generative metaphors” in the early stages of co-creation, and extends it concretely to the level of data encoding.

Inspired by this, we abandoned the unified threshold approach and shifted to personal calibration: each embroiderer embroidered a small sample using what she herself considered “sparse,” “medium,” and “dense,” serving as her personal density scale. The final interactive system was built to learn from these samples, creating an independent recognition model for each person. For colour encoding, we likewise adopted personal colour calibration cards, allowing the technical system to adapt to the individual colour variations inherent in hand-dyeing.

The epistemological significance of this repair process lies in the fact that the embroiderers’ craft knowledge calibrated the designers’ standardisation presuppositions, and the technical system, in turn, became the learner of handcraft differences. This constitutes an institutional response to the hermeneutical injustice identified by Fricker — not “accommodating” local knowledge within the existing technical framework, but using local knowledge to restructure the technical framework itself.

4.5 The Failure of Metaphor: Generational Divergence in Colour Negotiation

Not all metaphor negotiations proceeded so smoothly. When discussing how to use colour to represent pain intensity, another friction in this study emerged — this time among the embroiderers themselves.

Younger embroiderers favoured using “light blue” to represent the mildest pain, because it is “like a breeze blowing past, a little cool.” But a 68-year-old older embroiderer firmly objected: “When I was young, pain was burning hot, even the mildest was like fire. Your blue, it looks chilly to me — how does that look like pain?”

This is a case of metaphor failure. The younger embroiderers used a synaesthetic metaphor (mildness of pain ≈ coolness of wind), while the older embroiderer’s bodily experience tightly bound pain with “fiery heat,” making it impossible for her to accept “cool” as a legitimate representation of pain. From the perspective of Lakoff and Johnson’s theory, metaphorical mapping is highly dependent on individual experience — when two people’s bodily sensory frameworks for pain differ, the same metaphor cannot establish shared meaning between them.

In the end, light blue was retained as one option for the Level 1 colour, but the group agreed not to impose a unified sensory label for colour naming. Instead, each person was allowed to define the five-level colour correspondence according to her own silk thread colour spectrum during the personal calibration phase. This compromise effectively acknowledged that, in colour encoding, complete collective consensus might be unattainable, with personal calibration serving as a secondary safeguard.

The importance of this case lies in its illustration of the boundaries of metaphor as a negotiation tool. Metaphor can efficiently build consensus where shared bodily experience is ample (as in “embroidered dense means painful for long”), but when bodily experience itself differs significantly between individuals or generations, metaphor cannot bridge the rift. At the methodological level of co-design, this means researchers should not force the universalisation of metaphor, but should preserve individualised exits for it. This point is relatively underexplored in co-design literature — we tend to eagerly report the success of “generative metaphors” while remaining silent about their failures (see Akama and Light 2018 for a discussion of the constructive role of “discomfort” in participatory design).

4.6 Non-coercive Silence: The Dimension That Was “Skipped”

The negotiation on encoding labour type also merits recounting. We proposed using background textures (wave pattern = rice transplanting, grid = weaving, dotted lines = tea-picking) to assist annotation, and explicitly stated that this dimension was “non-mandatory.” Among the final 14 embroidered pieces, only 5 included background textures.

We initially interpreted this as a technical difficulty issue. But in follow-up interviews, one embroiderer who did not include texture said: “That day I picked tea, also transplanted rice, also wove. You do five or six things in a day — which one do you want me to embroider? Embroidering just one is incomplete, embroidering many makes a mess.” Another said: “It’s about telling where the body hurts anyway — doesn’t it hurt the same no matter what work you do?”

Their silence was not “not understanding” or “not having time,” but a proactive selective narrative. The distinction of labour type — a meaningful variable in our researcher framework — was, in their view, unnecessary for the core purpose of “telling the body’s story.” This non-use is essentially a silent recalibration of the researchers’ presuppositions: you wanted to record the division of labour; what we wanted to say is about the pain itself. This subtle moment reminds us that in co-design, participants’ “non-adoption” is just as important as “adoption”; silence is sometimes not absence, but a form of expression [5].

4.7 The Final Encoding Protocol: A Distributed Convention That Respects Difference

The encoding protocol that finally emerged was superficially similar to the researchers’ initial standardisation scheme, but its underlying logic was entirely different: it was a distributed encoding convention anchored by metaphor and cultural consensus, and realised through personal calibration.

5. Data Embroidery Pieces: The Materialisation of Bodily Knowledge

After the encoding rules were established, each woman independently created her personal embroidery piece based on her pain diary from the preceding week. The creative process itself became an embodied re-enactment — as Sister Pan said when recording her oral narrative: “When embroidering it, every stitch that went in felt like that day’s back pain came back again.”

The nature of this re-enactment merits deeper examination. It was not a simple recollection of pain, but a reactivation of bodily memory through the physical resistance of needle passing through cloth. Ingold (2013), in discussing the relationship between making and cognition, points out that the craftsperson does not first form a complete image in the mind and then execute it, but continuously generates form in response to the material’s resistance and feedback. What Sister Pan’s “every stitch felt like the pain again” reveals is the same process: encoding is not a symbolic operation external to pain, but an embodied revisiting of pain. It was in this revisiting that invisible bodily experience was sewn into material, acquiring a form that can be touched and preserved.

The 14 embroidered pieces ultimately presented 14 distinctly different visual styles. Under the same encoding rules, some pieces were compositionally full and colourfully intense, others sparse and light. This difference itself is information — it is not only the encoding of pain data, but also the unconscious imprint of personal temperament, aesthetics, and bodily state. In this sense, the embroidered pieces transcend the function of data carrier and become material witnesses to bodily narratives [9].

6. Discussion: Metaphor, Justice, and the Ethical Costs of Technology

After the embroidery pieces were completed, responding to the embroiderers’ wish for “more people to be able to read these body stories,” we co-developed an interactive system that allows viewers to touch the floral motifs on the embroidery pieces, triggering animations of bodily pain at that site alongside the maker’s oral narrative [8]. The technical core of the system is the personal calibration pipeline: the density scale samples and colour calibration cards of each creator were used to train her individual recognition model. This design embeds respect for individual handcraft difference into the technical architecture itself. The system achieved relatively high accuracy in external interpretation tests, but we will not elaborate on technical evaluation details here — because this paper is concerned with the epistemic politics of the co-design process, not system performance. Starting from the design logic of this system, we discuss three more fundamental issues.

6.1 The Efficacy and Boundaries of Metaphor as a Negotiation Tool

The empirical results of this study indicate that, in the co-design of cross-cultural data encoding, embodied metaphors spontaneously generated by participants can achieve a negotiation function that abstract technical language cannot. Among these, ontological metaphors (reifying abstract experience, such as “embroidered dense means painful for long”) had the highest consensus efficiency because they were built upon shared making-body experience. Morphological metaphors (like the flower-body site mapping) relied on visual similarity and existing cultural symbols, and were also relatively stable. Synaesthetic metaphors (like using colour temperature to express pain sensation), however, carried a risk of failure due to significant inter-individual differences in bodily experience — exactly what we observed in the generational colour divergence.

For co-design methodology, this implies that metaphor should not be romanticised as an all-purpose negotiation tool. It is extremely effective when its source domain has a broad basis of shared bodily or cultural grounding; but when the source domain itself (such as temperature perception, colour association) is markedly influenced by individual physiology and generational experience, metaphor needs to be supplemented by individualised exits. In other words, metaphor builds the consensus, personal calibration catches the difference — the combination of the two constitutes a more robust cross-cultural encoding negotiation mechanism than relying solely on either one.

Although the specific context of this study is embroidery craft and bodily pain data encoding, this finding holds transferable insights for broader co-design practices: in any situation involving negotiation across knowledge systems — for example, indigenous ecological knowledge and environmental science, traditional medical practice and public health, community spatial memory and urban planning — metaphor can serve as a more effective bridge than technical terminology. However, employing metaphor requires simultaneous clarity about its boundaries: when participants have markedly different bodily experiences of a metaphor’s source domain (such as temperature, colour, touch), consensus should not be forced. The designer’s role is not to select the “best metaphor” and promote it, but to identify the conditions under which metaphor is effective, and to preserve exits for individualised expression when metaphor fails.

6.2 From Hermeneutical Injustice to Institutional Response

Fricker’s concept of hermeneutical injustice reveals the structural predicament of marginalised groups lacking legitimate conceptual tools to interpret their own experiences. In this study, the embroiderers initially faced precisely such a situation: they possessed the experience of bodily pain, but lacked a tool that could translate this experience into a data form that could be recognised and traced by others.

Our co-design practice offers an institutional pathway for responding to hermeneutical injustice — not simply “providing tools for them,” but involving them in defining the encoding rules of those tools. When Sister Pan said “embroidered dense means painful for long” and formulated the encoding protocol on this basis, she was in effect creating an alternative hermeneutical resource: a data language rooted in craft epistemology, rather than in medical scales. The design of the personal calibration system further institutionalised this alternative hermeneutical resource — it compels the technical system to learn each person’s handcraft “dialect,” rather than conversely demanding that people learn the machine’s standardised language.

However, we must also acknowledge the limits of this response. As Harrington, Erete, and Piper (2019) discuss in CoDesign regarding participatory health research, data tools co-created by communities, even if internally recognised, still face uncertain legitimacy within formal medical systems. The embroidered pieces can currently only be “read” within the community and in the researcher-built interactive system. Between institutional response and institutional recognition lies a long and uncertain road. This limitation is not a failure of this study, but a reality that any project attempting to address structural injustice through co-design in marginalised communities must face — institutional change is not the output of a design project, but a process that requires synergistic advancement with policy, medical systems, and long-term community action.

Data Dimension	Encoding Element	Encoding Scheme	Negotiation Mechanism	Tensions / Unresolved
Pain Intensity	Thread Colour	Five-level colour scale, personal calibration for specifics	Synaesthetic metaphor partially successful; generational divergence bridged by personal calibration	Young/old differing experiences of “cool-coloured pain” not unified
Duration	Stitch Density	Sparse, Medium, Dense, intervals defined by personal calibration card	Ontological metaphor “embroidered dense means painful for long” reached high consensus	Cross-individual comparison difficult; this incommensurability intentionally preserved
Body Site	Floral Motifs	Peony, maple leaf, lotus, plum blossom, chrysanthemum	Morphological metaphor and cultural symbol consensus, highly smooth	Almost no dispute
Labour Type	Background Texture	Four textures, non-mandatory	Researcher-proposed, only partially adopted	9 people actively chose non-use, a collective negation of this dimension’s narrative necessity

7. Conclusion

This paper reports an epistemological collision and its repair in data embroidery co-design — and the new problems exposed after the repair. From a methodological perspective, the main contribution of this study lies not in producing an encoding system, but in revealing three interlinked mechanisms within the co-design process.

First, the efficacy and boundaries of metaphor as negotiation infrastructure. Through the successful case of “embroidered dense means painful for long” and the failure case of generational colour divergence, we have illustrated the potential and limits of embodied metaphor in cross-cultural encoding negotiation, offering a more nuanced argument for co-design methodology than simply “metaphor is useful.”

Second, the mechanism by which craft epistemology recalibrates technical rationality. Sister Pan’s refusal was not a negation of the proposal, but a fundamental challenge to its epistemological premises. This collision forced us to twist the technical system from the direction of “unified standard” toward the direction of “learning individual difference” — an attempt, at the level of system design, to respond to hermeneutical injustice.

Third, the self-correction of the researchers’ epistemological presuppositions. The embroiderers’ polite correction of our over-theorising reminded us that power asymmetry in co-design manifests not only at the institutional level, but also at the level of knowledge production. Researchers, too, can, in the name of “understanding,” translate participants’ improvisational expressions into academic language and claim ownership of them. Genuine epistemic humility requires us to continuously expose, rather than hide, this risk throughout the research process — this reflection is not a one-time epiphany, but a process of repeatedly slipping into and being pulled back from abstraction throughout research and writing.

It must be acknowledged that this paper has several limitations. The study was conducted in a single village and with a specific craft tradition; the transferability of the encoding rules and methodological findings requires testing in multiple contexts. We were unable to track the project’s long-term impact on community power relations, nor to adequately assess the potential exclusionary effects of the personal calibration scheme. The long-term sustainability of the technical system remains an unresolved question. Moreover, this study did not track the actual impact of the co-design process on participants’ physical health conditions, healthcare-seeking behaviours, or intra-family pain narratives [10]. This gap limits our definition of the project’s “success” — whether it merely achieved visibility at the narrative level, or whether it can generate substantive effects at the level of health empowerment, awaits answers from long-term follow-up research. These limitations are not failures of the study, but reflect the inherent openness of co-design itself — it is never a project that can be “completed” and “delivered,” but a continuous, uncertainty-laden process of negotiation.

For researchers working in similar contexts in the future, we suggest: do not enter a community with a set of “better encoding standards,” but create a co-design space where local knowledge can calibrate your own presuppositions; when participants spontaneously generate metaphors, recognise and cherish them, but do not force their universalisation, and do not rush to theorise them — they are first the participants’ knowledge, and only then can they become nourishment for theory; when designing technical solutions, treat individual difference as a cultural resource rather than technical noise, but at the same time candidly face its ethical costs — cross-individual incommensurability, sustainability burden, and potential exclusion — and incorporate an “exit plan” into the co-design agenda from the very beginning. The “co-” in co-design does not mean researchers and participants drawing the same picture at the same table, but rather, within the looseness and tightness of their own stitches, each embroidering their own bodily truth — and then letting them be seen, be touched, be respected, but not reduced to a single standard.

References

- [1] Trifunovic, B. (2025) Reshaping Cultural Preservation, Digital Innovation, and Technological Advancement PDT&C 54-4 Editorial. *Preservation, Digital Technology & Culture*, Vol. 54 (Issue 4), pp. 259-261. <https://doi.org/10.1515/pdtc-2025-0075>
- [2] Xu, Q., Zhou, Z., Cao, N. and Liu, G., 2025. From Struggle to Enjoyment: Investigating ICH Digital Inheritance Through the Lens of Vocational Education. In: *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 1, 2025, Yokohama, Japan. New York: ACM, pp. 1–7. doi:10.1145/3706599.3719992
- [3] Liu, Z., Sun, H., Peng, Z., Duan, Z. and Pan, D., 2024. Cultural transition by digital technologies: invasion or empowerment? In: C. Gray, E. Ciliotta Chehade, P. Hekkert, L. Forlano, P. Ciuccarelli and P. Lloyd, eds., *DRS2024: Boston*, 23–28 June 2024, Boston, USA. London: Design Research Society. doi:10.21606/drs.2024.976
- [4] Tunstall, E., 2023. *Decolonizing Design: A Cultural Justice Guidebook*. Cambridge, MA: MIT Press.
- [5] Jovicic, S., Göbl, B., Verstappen, S. and Kayali, F., 2025. Designing amidst uncertainty: drifting and byproducts in the intersection of anthropology and computer science. *CoDesign*, 21(2), pp.308–323. doi:10.1080/15710882.2024.2396945
- [6] West, P., 2020. Designing in response to Indigenous sovereignties. In: *Proceedings of the DRS 2020 International Conference: Tensions, Paradoxes, Plurality*, 11–14 August 2020, Melbourne, Australia. London: Design Research Society. doi:10.21606/drs.2020.319
- [7] Petkova, K., 2025. Community cultural centers and the digital divide: between local engagement and systemic constraints. *KIN Journal*, 11(2), pp.1–14. doi:10.55630/KINJ.2025.110213
- [8] Lalopoulos, A., 2025. Digital storytelling for intangible cultural heritage: a mobile application for the Museum of Asia Minor Culture. In: *Proceedings of the CAA 2025 International Conference*.
- [9] Giglitto, D., Ciolfi, L., Lockley, E. and Kaldeli, E., eds., 2023. *Digital Approaches to Inclusion and Participation in Cultural Heritage: Insights from Research and Practice in Europe*. Abingdon: Routledge. doi:10.4324/9781003277606
- [10] Bavi, A., 2025. *Seeing Through Touch: Creating Inclusive Experiences for Blind and Partially Sighted Museum Visitors Through 3D Printing*. PhD thesis. University of British Columbia (Okanagan).



Embroidery as Language.
Bodies as Archives.
Data as Shared Stories.



PORTFOLIO
of Practice



Lanzhou Wales College
School of Art and Design
Practice-based Research